Food Engineering Reviews (2025) 17:75–103
https://doi.org/10.1007/s12393-024-09388-0

AI-Enabled Optical Sensing for Smart and Precision Food Drying: Techniques, Applications and Future Directions

Marcus Vinicius da Silva Ferreira[1] · Md Wadud Ahmed[1] · Marciano Oliveira[1] · Sanjay Sarang[2] · Sheyla Ramsay[2] · Xue Liu[3] · Amir Malvandi[1] · Youngsoo Lee[4] · Mohammed Kamruzzaman[1]

Received: 19 October 2024 / Accepted: 11 November 2024 / Published online: 20 November 2024
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2024

Abstract
Recent developments in alternative drying techniques have significantly heightened interest in innovative technologies that improve the yield and quality of dried goods, enhance energy efficiency, and facilitate continuous monitoring of drying processes. Artificial intelligence (AI)-enabled optical sensing technologies have emerged as promising tools for smart and precise monitoring of food drying processes. Food industries can leverage AI-enabled optical sensing technologies to gain a comprehensive understanding of drying dynamics, optimize process parameters, identify potential issues, and ensure product consistency and quality. This review systematically discusses the application of selected optical sensing technologies, such as near-infrared (NIR) spectroscopy, hyperspectral imaging, and conventional imaging (i.e., computer vision) powered by AI. After covering the basics of optical sensing technologies for smart drying and an overview of different drying methods, it explores various optical sensing techniques for monitoring and quality control of drying processes. Additionally, the review addresses the limitations of these optical sensing technologies and their prospects in smart drying.

Keywords Smart drying · Optical sensing · Near-infrared · Hyperspectral imaging · Machine learning · Artificial intelligence · Chemometrics

Introduction

Drying is widely used to preserve food by reducing moisture content and enhancing shelf life. This centuries-old preservation method has gained significant importance in modern times due to its effectiveness in extending the shelf life of various food items [1]. Drying can alter taste, flavor, texture, and nutritional value, including vitamins and bioactive compounds, depending on specific drying conditions. Small changes in drying conditions sometimes significantly impact sensitive components, such as vitamin C and other thermolabile compounds [2, 3]. Therefore, to minimize the quality loss of dried products, demand for alternatives to conventional drying systems has increased [4]. Implementing advanced drying techniques for optimal quality would greatly benefit from integrating cutting-edge systems, particularly "smart systems," aimed at enhancing efficiency compared with conventional drying methods [5–7]. The concept of "smart" or "intelligent" drying involves utilizing advanced sensing techniques, data analytics, and advanced decision-making tools to automate the drying process and enable precise control of environmental conditions and product quality, ensuring optimal outcomes such as real-time monitoring of moisture levels. For example, smart integrations may include monitoring the drying rate and assessing the quality parameters of the dried food during drying to enhance drying efficiency and reduce energy consumption [7]. Furthermore, smart systems facilitate the automation and optimization of the drying process, reducing manual interventions [8].

Optical sensing technologies such as NIR spectroscopy (NIRS), hyperspectral imaging (HSI), and RGB imaging (also called computer vision-CV), offer diverse capabilities

✉ Mohammed Kamruzzaman
mkamruz1@illinois.edu

[1] The Grainger College of Engineering, Department of Agricultural and Biological Engineering, University of Illinois Urbana-Champaign, Urbana, IL 61801, USA

[2] PepsiCo R&D, Plano, TX 75024, USA

[3] BASF Corporation US, Beachwood, OH 44122, USA

[4] Department of Biological Systems Engineering, Washington State University, Pullman, WA 99164, USA

of capturing chemical, morphological, and visual data, applicable across fields including agriculture and food industries [9–12]. These optical sensing techniques have emerged as valuable, rapid, and non-destructive tools for process analysis, extensively used in food processing and quality control applications. They offer significant advantages, require minimal or no sample preparation, and are environmentally friendly, thereby eliminating the generation of chemical waste [9, 12]. For example, NIRS enables monitoring of moisture content and other components of the material being dried while HSI facilitates the monitoring of composition changes and their distribution and has found widespread application in various contexts of the food industry [9, 10]. CV can be used with image processing techniques to extract quantitative information about drying materials, including size, shape, shrinkage, and texture [13].

Considering smart integration, this review explores various optical sensing technologies such as NIRS, HSI, and CV with AI-driven machine learning (ML) and chemometrics in food drying systems. This study comprehensively assesses various drying technologies, identifying optimal optical sensing and ML approaches for each method and addressing challenges in industrial food drying processes. It further evaluates the advantages and limitations of AI-enabled optical sensing in drying systems. Finally, analyzing future demands and opportunities for AI-enabled optical sensing for food drying, this review provides a clear direction for future research and development.

Industrial Drying Methods

Industry dryers, traditionally comprising rotary, fluidized bed, drum, spray, freeze, microwave, and dielectric drying systems, are used to meet diverse requirements, and are engineered to efficiently remove moisture from food products. These drying processes can be broadly classified into batch and continuous drying, each with applicable contexts [14]. Batch drying involves processing the product in groups or quantities, making it suitable for smaller-scale operations such as commercial food service or kitchens. In contrast, continuous drying processes the product constantly, which is ideal for larger, more uniform production needs. The following section will discuss some selected industrial dryers, highlighting their characteristics and applications in detail.

Rotary Drying

Rotary drying plays a crucial role across industries due to its efficiency, versatility, and uniformity in drying materials. This method uses a rotating drum or cylinder to transport heat effectively and evaporate moisture. Due to its capacity to retain, improve, or modify a material's properties while reducing its moisture level, the technique is widely used in agriculture, food processing, chemical manufacturing, and mineral processing. They are particularly suited for handling large quantities of solid, granular, or somewhat chunky materials (grains, seeds, nuts, bio-solids). The rotary drying technique is valued for its ability to handle various materials with different sizes, shapes, and qualities. The rotary dryer simulation model was developed as early as the 20th century and employed as the cornerstone for formulating a practical computer control strategy. Methodologies for this drying type have been developed and thoroughly accessed in different frameworks [15]. Figure 1 represents the basic features of the rotary dryer.

The drum diameter (D) and length (L), drum slope (α), rotational speed (N), lifting flight number (nf), profile, drying gases direction (concurrent or countercurrent), and velocity (V) through drum length are among the key parameters that influence efficiency and effectiveness of the drying process. These parameters are used to calculate solids holdup H and outlet moisture content X_{out}. Heat transfer from hot gases to particles causes the moisture inside the particles to vaporize due to dynamics and particle movements by cascade and kiln motion through the drum length. The moisture finally mass-transfers from the interior of the particles to their surface, followed by heated gases in the drum. The heat transfer coefficient is defined as follows, where K is the mass transfer coefficient ($kg/m^2 \cdot s$), L is the length of the dryer (m), D is the diffusivity of moisture in solid (m^2/s), G is gas flow ($kg/m \cdot s^2$), and ΔT is the temperature difference (°C).

Fluidized Bed Drying

Fluidized bed dryers (Fig. 2) are essential for drying wet solid particles in industrial processes, though they face challenges like scaling up, poor fluidization, and varying product quality. Scaling up is challenging due to changing bubble behavior and mixing in larger dryers, and the lack of reliable models requires expensive testing. Conventional methods struggle to capture the complex dynamics, requiring innovative approaches for improved predictive capabilities. Fluidized bed drying involves suspending solid particles in a rising gas stream, creating a fluid-like state that enhances heat and mass transfer. Drying particulate materials, such as gas and particles, and overheating are controlled, which spares sensitive products. Another important aspect is that heat transfer rates between fluidized bed-immersed objects are increased. It is also easier to control the temperature within the process due to the fast mixing of solids, which enables isothermal conditions throughout the process [16].

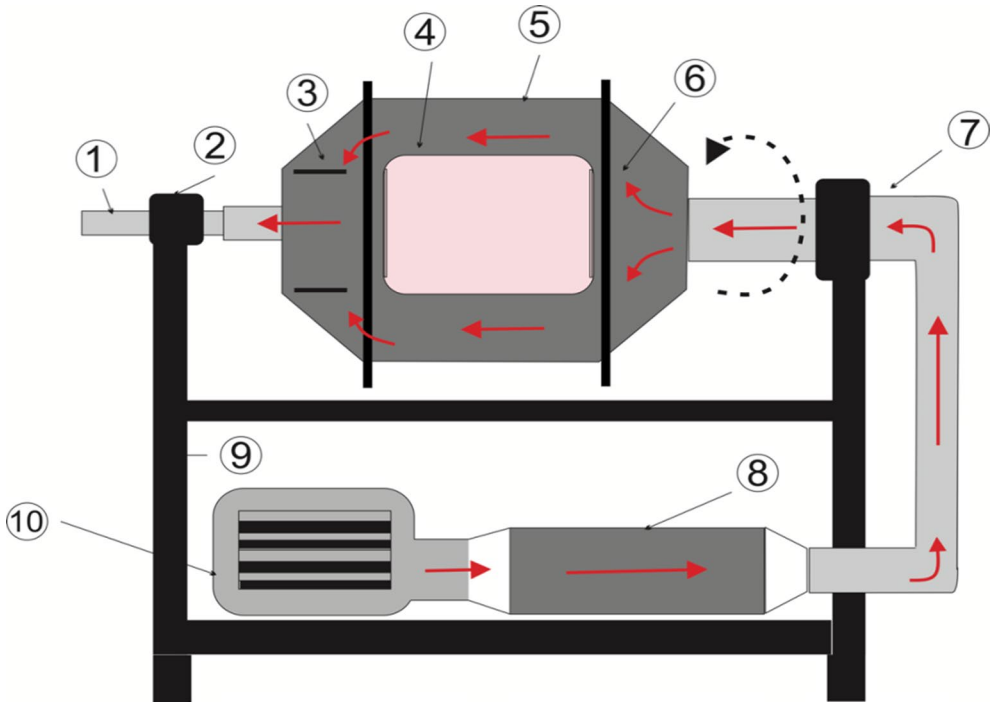
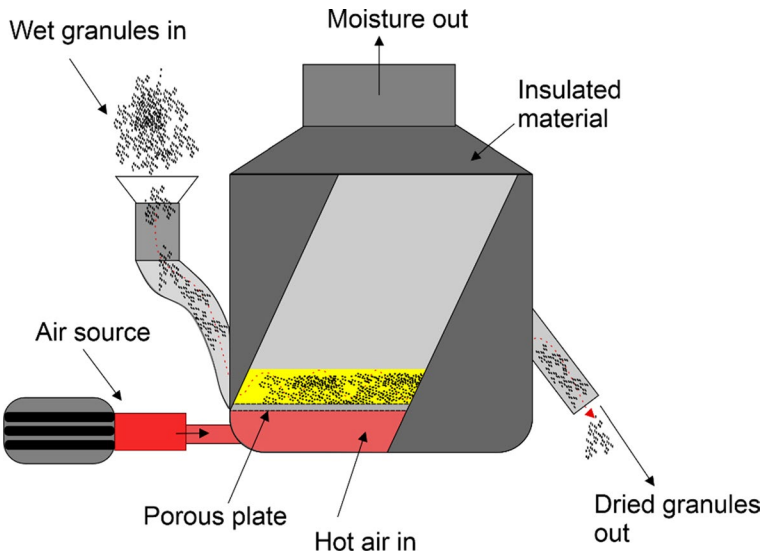


Fig. 1 Schematic diagram of a typical rotary dryer (**1**) exhaust pipe, (**2**) bearing, (**3**) exhaust chamber, (**4**) drying chamber, (**5**) rotary drum, (**6**) plenum chamber, (**7**) thermo-couple, (**8**) heating chamber, (**9**) mainframe, and (**10**) blower

Fig. 2 Schematic diagram of a typical fluidized bed dryer



Freeze-Drying

In freeze drying (Fig. 3), the product is initially frozen, then the ice is eliminated through sublimation, going straight from the solid to the vapor phase. The sublimation of ice during the freeze-drying process significantly alters the shape and volume of foods. The sublimating ice crystals produce pores or voids with various properties depending on the process parameters. Investigating the effects of freeze-drying process parameters on the structural characteristics of food products becomes increasingly intriguing Dehydrated products' structural characteristics, including elements like density and porosity, are linked to the texture and overall quality of the finished product. These structural characteristics significantly influence both taste and aesthetic attractiveness. Beyond porosity, a crucial factor in establishing these qualities is the distribution of pore diameters. It is possible to estimate pore size distribution accurately using image analysis of two-dimensional pictures [17].

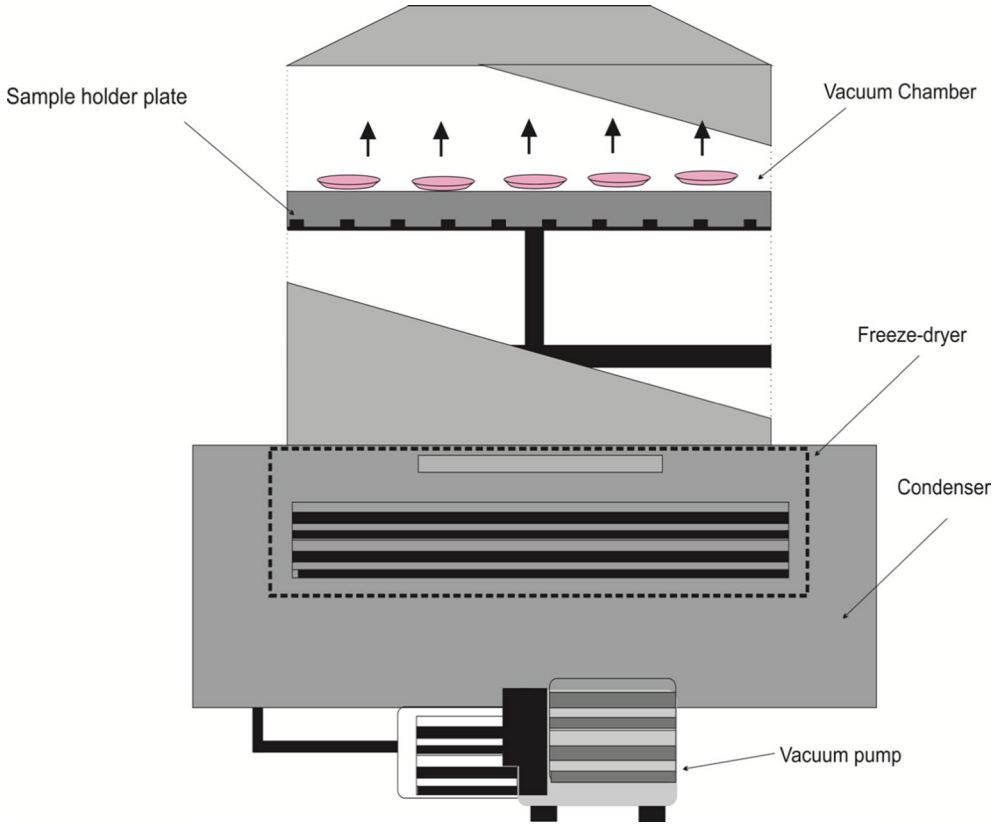


Fig. 3 Schematic diagram of a typical freezing unit

Spray Drying

In drying technologies, significant progress has been made with new approaches to better preserve thermosensitive and oxidizable compounds. Among these, two prominent techniques have emerged: vacuum and spray drying. Vacuum drying, which operates under reduced pressure, allows for the gentle evaporation of moisture and volatile components at lower temperatures, typically around 30 °C, thereby ensuring the integrity of heat-sensitive materials [18]. On the other hand, spray drying (Fig. 4**)** involves atomizing suspensions in a drying chamber, facilitating rapid vaporization due to increased surface area exposure. Notably, combining these techniques has led to the development of vacuum spray drying- an innovative process that combines the benefits of both vacuum and spray drying methodologies. Extensive studies [18, 19] have demonstrated the efficacy of vacuum spray drying, particularly for thermosensitive compounds, even when external heat sources are utilized to provide latent heat for evaporation. Moreover, the absence of airflow in vacuum spray drying offers additional protection to oxidizable compounds, further enhancing its utility in diverse applications.

Microwave Drying

Microwave drying (Fig. 5) rapidly removes moisture from materials using microwave radiation, renowned for its efficiency and widespread application in the food industry. Microwave drying enhances product quality by minimizing sample shrinkage and preserving surface moisture, particularly in the falling rate phase, as evidenced by its effectiveness for different samples, such as berry fruits [20], ginger [21], and carrots [22]. Hybrid methods like microwave-vacuum drying improve efficiency and quality using lower boiling points in vacuum conditions, as demonstrated by [23] during the drying of sea cucumber, which resulted in minimal loss of amino acids and fatty acids compared to traditional drying techniques.

Infrared Drying

Infrared radiation (IR), electromagnetic waves between 0.75 and 1000 mm from a heating source, warms the product's surface directly, not the surrounding air. As shown in Fig. 6, IR drying is used for various crops due to its fast-drying speeds and low risk of burning, effectively preserving total phenolic content better than other methods. Its advantages include high energy efficiency, reduced drying times, and

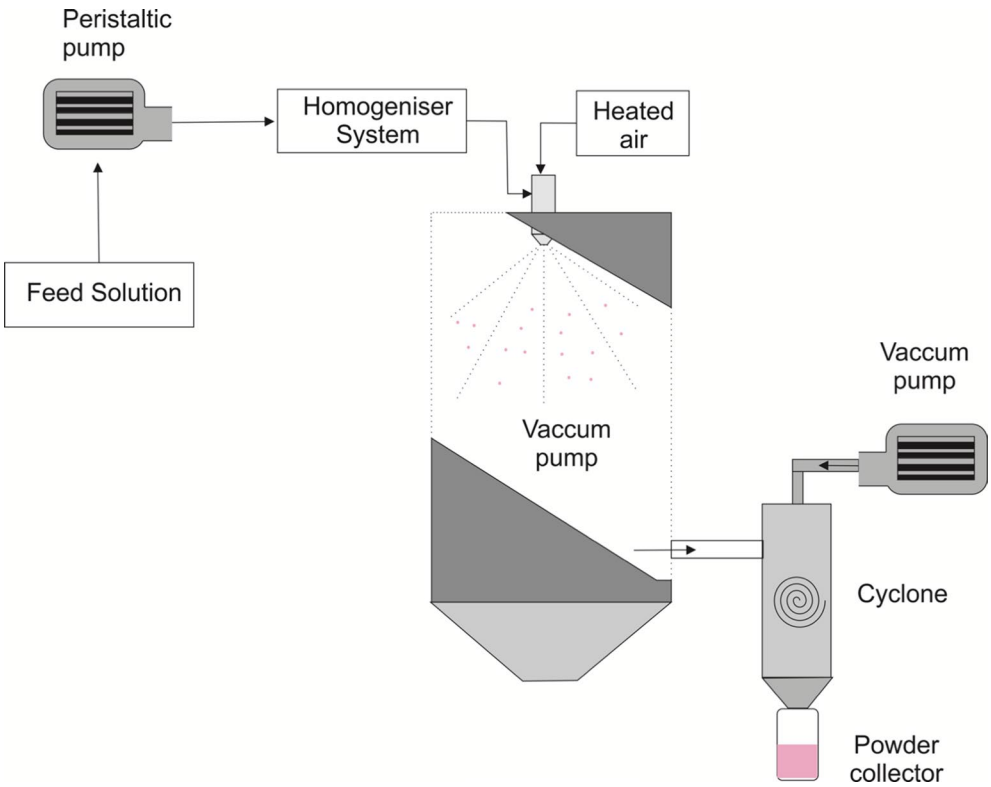


Fig. 4 Schematic diagram of vacuum-spray drying system

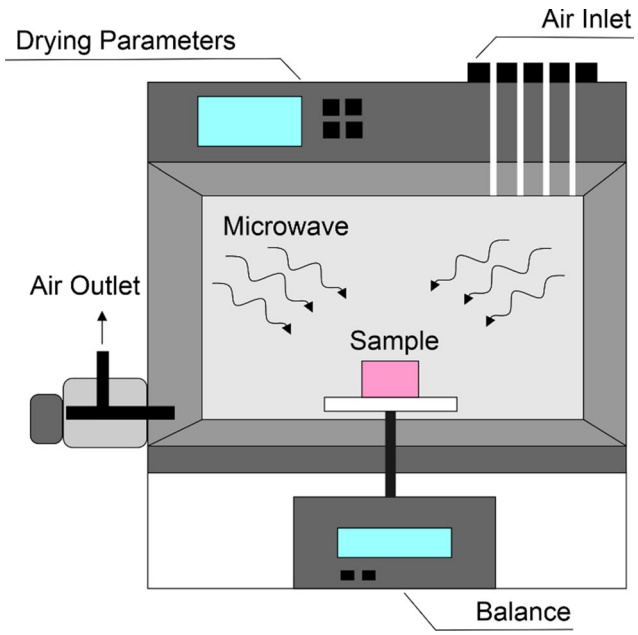


Fig. 5 Schematic diagram of a microwave-convection oven dryer

straightforward temperature management, highlighting its potential for food products over conventional heating systems [25, 26]. IR heating has been applied in drying, baking, roasting, blanching, pasteurizing, and sterilizing food products [25].

Convective Hot-Air Oven Drying

Convective continuous hot-air dryers, commonly utilized in industrial production lines, are crucial in enhancing final product quality (Fig. 7). Despite their significance, challenges regarding energy consumption, complex thermophysical processes, and monitoring difficulties within the oven. With the rising energy cost, there is a growing emphasis on enhancing thermal transfer efficiency in these ovens [27, 28]. Convection ovens are typically structured in zones, each featuring a single heater and circulation fan. The fan propels air through ducts, distributing hot air onto products via slots or nozzles above and below the baking band. The airflow is then drawn back for recirculation through the system, with each zone equipped with extraction fans and flues to eliminate moisture from the baking chamber. Control over the baking process primarily relies on temperature modulation and adjusting the airflow distribution [28].

Various approaches are employed to evaluate and predict oven performance to mitigate processing time and boost thermal efficiency. These methods include experimental, theoretical, and computational studies. While full-scale experimental approaches yield the most accurate results, they are costly and time-consuming. Theoretical methods, relying on mathematical models, are limited to simplified cases and struggle with complex phenomena. Computational

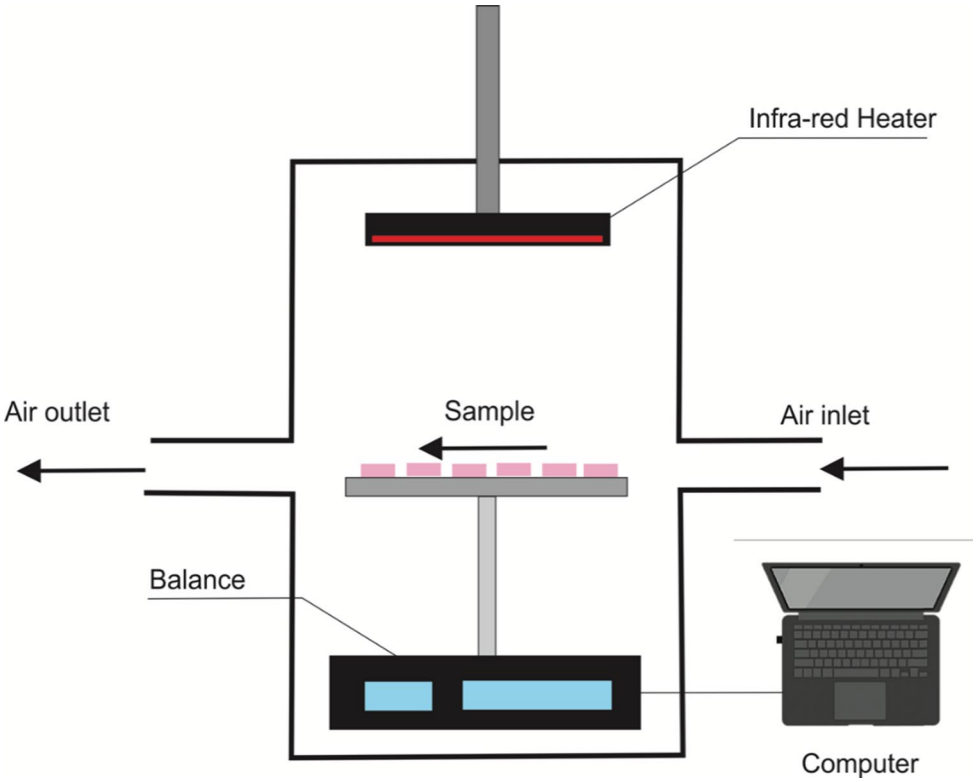


Fig. 6 Schematic diagram of Infra-red drying equipment

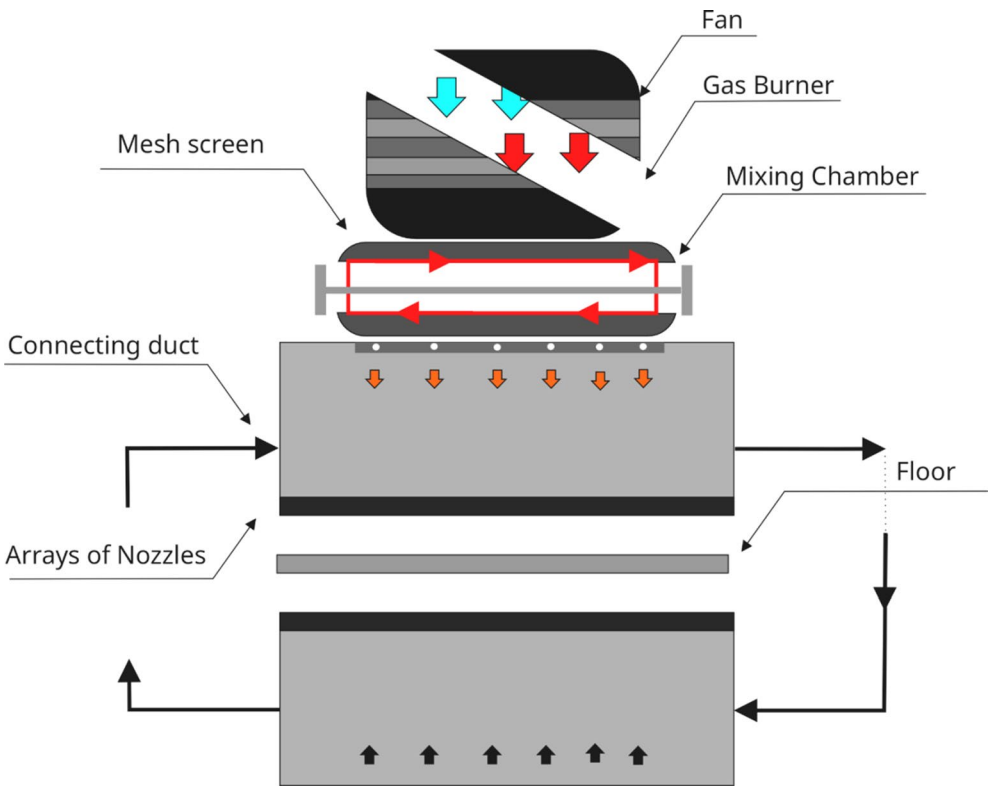


Fig. 7 Schematic diagram of a convective continuous hot-air dryer (industrial oven)

methods employing tools like Computational Fluid Dynamics (CFD) offer benefits such as reduced lead time and cost during the design stage, particularly for assessing large and intricate systems [27]. Experimental and theoretical research has been conducted to optimize drying parameters and energy characteristics in industrial convection dryers with pneumatic material transport. This research provides valuable insights for designing and constructing such dryers in the food industry, focusing on technical specifications, energy balances, and heat transfer coefficients [29]. Khatir et al. [30] highlight using two-dimensional CFD modeling to assess temperature profiles and airflow patterns within bread-baking ovens.

Hot air drying has been used alone and in combination with microwave and infrared for many crops such as fruits, vegetables, and seeds [31] and meat products [32].

Smart Drying Systems

A cutting-edge method known as smart drying systems (SDS) uses sensors to monitor and control several quality criteria while continuously drying. This advanced technology uses ML algorithms to optimize performance, promote environmental responsibility, and conserve energy. By monitoring humidity levels and moisture content, these systems offer better accessibility and control. Users can leverage remote connectivity to oversee drying processes from anywhere, enhancing operational flexibility and efficiency [33]. Ongoing research progress has driven the industry to innovate quality control within drying technology. The accessibility of smart dryers has increased in recent years, initially in laboratory-scale investigations and later in industries necessitating real-time food quality control [34].

Fundamentals of Optical Sensing Technology Used for Smart Drying Systems

Optical sensing technologies such as NIRS, HSI, and CV enable real-time moisture content and chemical composition monitoring during drying processes, facilitating precise control and optimization. These technologies are briefly described in the following sections.

Near-Infrared Spectroscopy

NIRS, a powerful form of high-energy vibrational spectroscopy, operates within the wavelength range of 700 to 2500 nm, with its utility rooted in the absorption of electromagnetic radiation by molecular vibration associated with C-H, O-H, and N-H chemical bonds. The resulting NIR spectra reflect a sample's physical and chemical properties, displaying broad bands due to overlapping absorptions corresponding mainly to overtones and combinations of vibrational modes. A typical NIRS system is shown in Fig. 8.

As NIR radiation interacts with a sample, its spectral characteristics undergo changes influenced by wavelength-dependent scattering and absorption processes. These alterations depend on the sample's internal chemical composition, such as moisture content, light scattering properties, and particle size. Despite the potential for background noise and overlapping bands in NIR spectra, ML methods, including multivariate analysis or chemometric techniques, prove effective in developing relationships between spectra and selected quality or compositional parameters [35].

NIRS's ability is particularly evident in monitoring drying rates and moisture content, leveraging the strong absorption bands of water in the NIR wavelength region. Critical absorption bands at 1440 nm (the first overtone of OH stretch) and 1940 nm (a combination of OH bending and stretching) facilitate detailed insights into the water molecule structure. The technique has several advantages, including a longer penetration length, up to 10 mm in the short wavelength region (750–1100 nm). This makes studying intact biological samples a rapid and non-destructive measurement technique [36].

Widely recognized for its applicability, NIRS has proven invaluable in determining moisture content and analyzing quality composition in food matrices. The strong C-H, O-H, and N-H absorption bands in NIR spectra make it an effective tool for simultaneously analyzing multiple attributes. Emphasizing its technic-economic advantages and versatility, NIRS is one of the most promising technologies for on-/in-line monitoring and control of processes and product quality in the food processing industry [37–39]. Over the past decades, continuous development has elevated NIRS to a fast and robust non-destructive technology, contributing significantly to its widespread adoption across diverse scientific and industrial domains.

Hyperspectral Imaging

HSI transforms drying processes across various industries by integrating imaging and spectroscopy in one system. This advanced optical sensing technology offers detailed spectral information for each pixel in an image, enabling unprecedented analysis and control. Currently, HSI is the only technology that answers the critical analytical questions of what is present, how much, and where it is located. Exploring the applications of HSI in smart drying reveals its potential to enhance efficiency and product quality optimization.

Core Components of Hyperspectral Imaging Systems An HSI system consists of vital components for capturing and

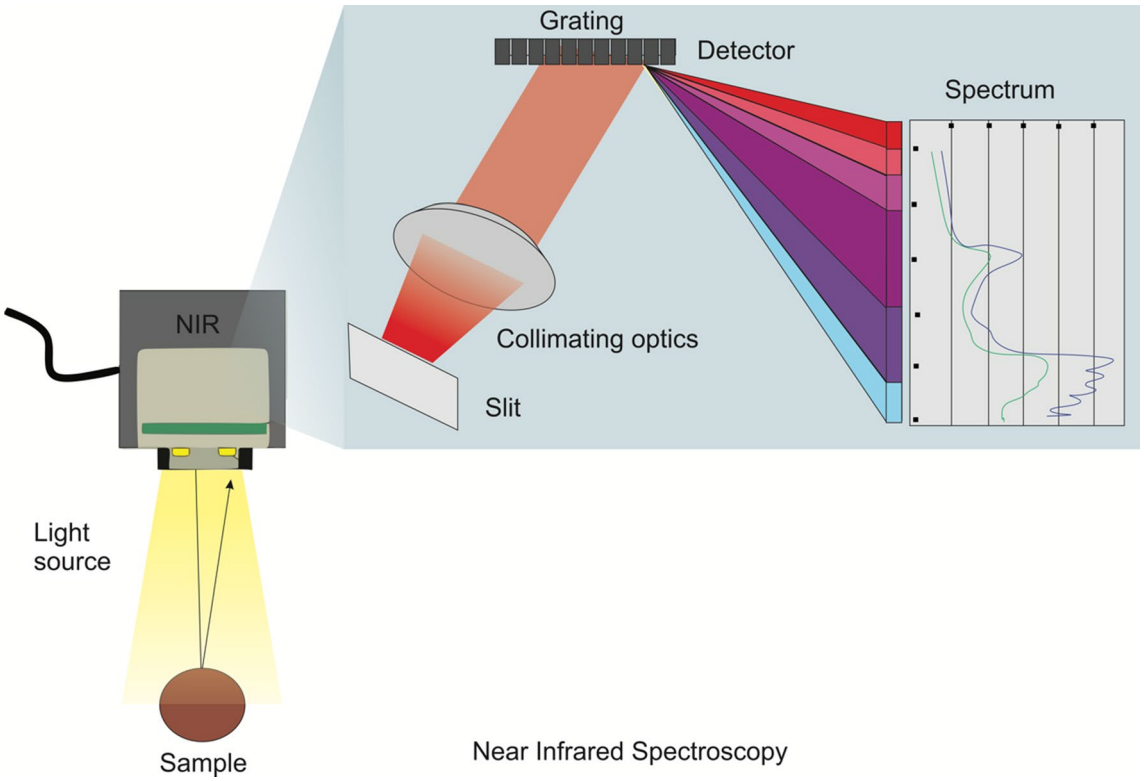


Fig. 8 Schematic diagram of a typical NIRS equipment

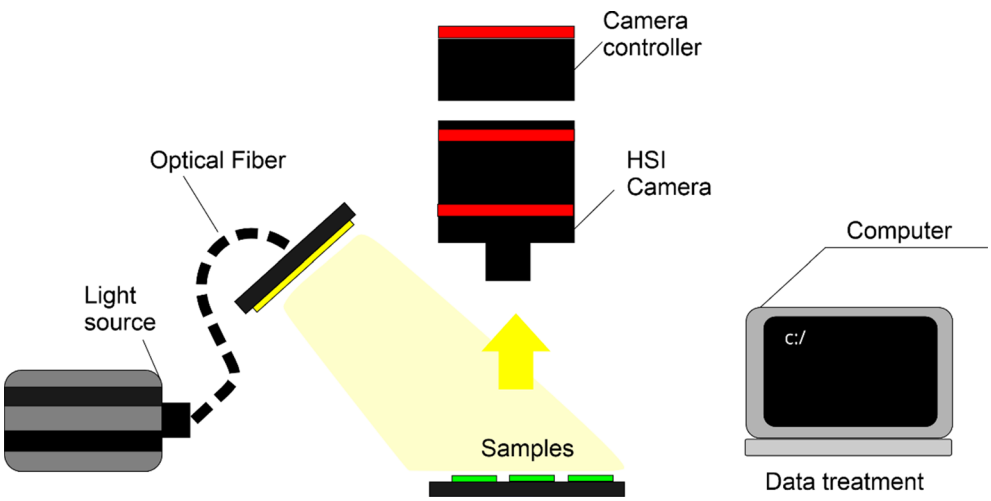


Fig. 9 Schematic diagram of a typical HSI system

analyzing intricate spectral details. These include a light source, detector, optical elements, and a computer with software for image acquisition, digitization, storage, modeling, and decision-making processes (Fig. 9). The light source serves as a fundamental element, with HSI systems employing various sources such as light-emitting diodes, halogen lamps, tunable light sources, and lasers to ensure optimal sample visibility during imaging. Wavelength dispersion devices, including liquid crystal tunable filters, acousto-optical tunable filters (AOTFs), filter wheels, Fourier-transform imaging spectrometers, imaging spectrographs, and single-shot imagers, play a pivotal role in dispersing broadband light into distinct wavelengths, enabling detailed spectral analysis. Area detectors, predominantly solid-state detectors like charge-coupled devices (CCD) and complementary metal-oxide-semiconductor (CMOS) cameras, are essential for quantifying light intensities. Equipped with light-sensitive photodiodes, these detectors convert incident

photons into electronic signals, facilitating the precise capture and processing of hyperspectral data for various applications [40].

Techniques in Hyperspectral Imaging In HSI, imaging or scanning methods encompass point, line, and area scanning techniques, each offering unique advantages and applications. Point scanning involves capturing spectral information from individual points on the sample, providing high spatial resolution but requiring longer acquisition times. On the other hand, line scanning involves scanning along a single line or path across the sample, offering faster imaging speeds while sacrificing some spatial resolution. Area scanning covers the entire sample surface, providing comprehensive spectral data with moderate spatial resolution. These scanning methods can be tailored to specific applications, balancing resolution, speed, and coverage to achieve optimal results in various fields, including remote sensing, agriculture, biomedical imaging, and material analysis [41, 42].

Computer Vision

CV plays a crucial role in monitoring the drying process of materials by providing a non-invasive and real-time method to observe changes in physical properties. High-resolution images captured over time can reveal variations in color, texture, and shape, indicative of moisture content and drying rate. These visual signs enable precise tracking of drying stages, ensuring optimal conditions are maintained to achieve uniform drying. Additionally, CV allows for detecting defects, such as cracks or uneven drying, which can compromise the quality of the final product. By integrating image analysis techniques, such as edge detection and pattern recognition, traditional imaging enhances the accuracy and efficiency of the drying process, supporting better control and optimization in various industrial and agricultural applications [43–45]. A typical CV or RGB system is shown in Fig. 10.

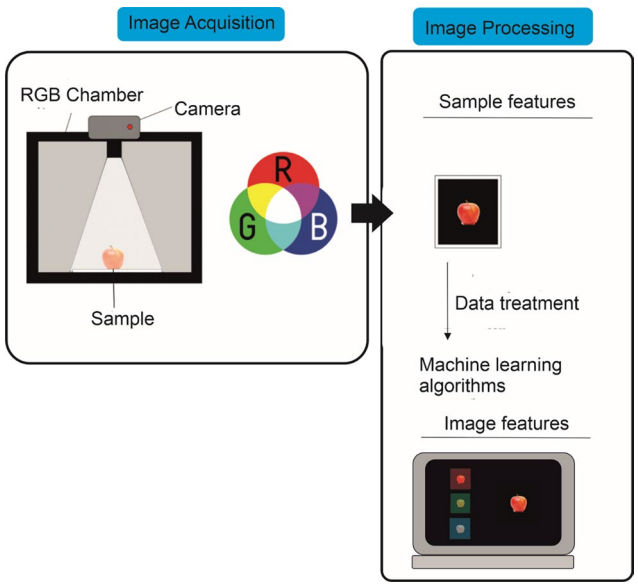


Fig. 10 Schematic diagram of a typical RGB system setup

Other Optical Sensing

Other optical sensors can be used with SDS, including thermal imaging, infrared thermography (IRT), and Terahertz imaging (TI). A thermal imaging system is shown in Fig. 11. The infrared detector captures emitted radiation from samples and converts it into electrical signals, which is crucial for accurate assessment. By measuring the infrared light emitted, infrared thermography determines the temperature of a surface. Detector sensitivity, assessed using blackbody radiation principles, directly impacts system performance [24]. It may be used to optimize and regulate the temperature conditions by keeping an eye on heat dispersion during the drying process. TI may offer details on materials' composition and water content by penetrating them using terahertz frequencies [45]. It is very helpful for monitoring and analyzing the water content of different compounds without causing damage. While commercially available options like NIR, NIR-HSI, and RGB offer robust advantages, IRT and TI have also been extensively explored in the food processing industries [25, 45]. Therefore, this review will be limited to NIR, HSI, and RGB optical-driven technology.

AI-Driven Optical Sensing for Smart Drying

In smart drying processes, AI has shown significant practical benefits in industrial applications. Notable contributions have demonstrated that AI enhances productivity, reduces costs, and improves product quality in food drying. Khan, Byun, and Park [46], along with Kushwah et al. [47], highlight AI's key role in developing technologies for precise selection of properties like color, shape, texture, volume, and weight, as well as optimal control parameters such as drying time, air temperature, airflow, moisture content, and final thickness. These AI-driven advancements have modernized food processing, improving the yield and quality of various food products. Furthermore, Zhu et al. [48] and Santhi and Muthuswamy [49] emphasized the relevance of incorporating AI in food drying technology from scientific and industrial perspectives. AI addresses real-time challenges by enabling specific product parameter monitoring and quality control throughout manufacturing, leveraging cloud-based information storage for continuous accessibility.

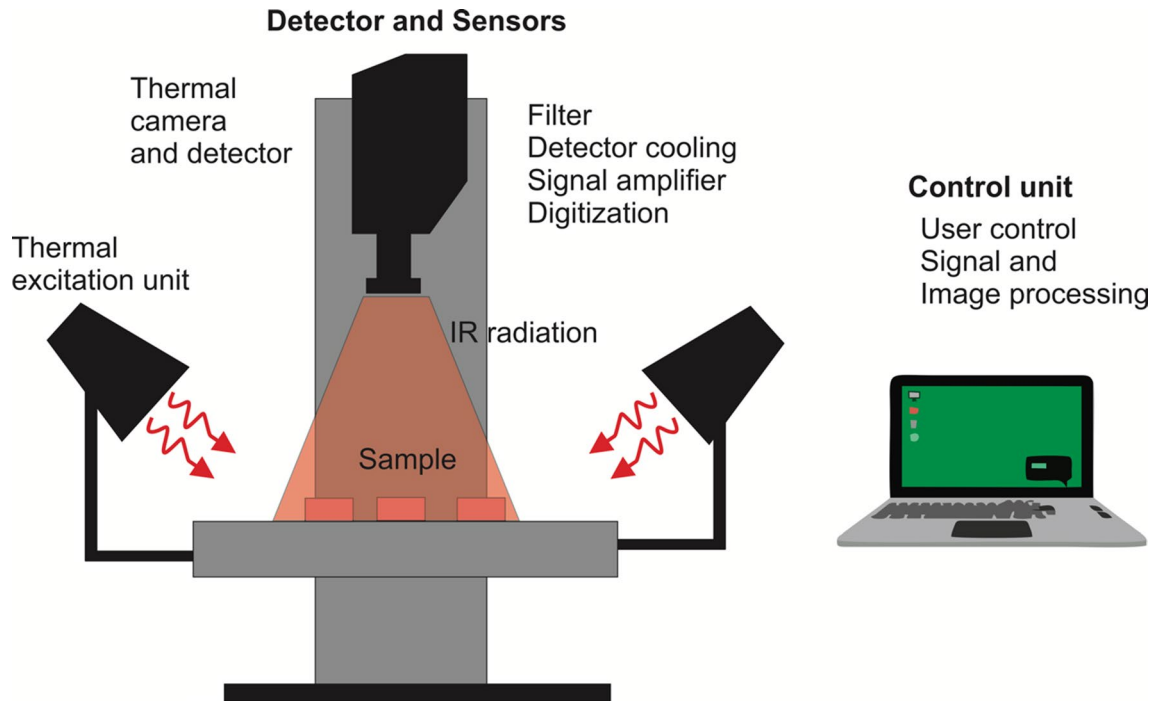


Fig. 11 Common elements found in a thermal imaging setup utilized for evaluating the quality of food materials

ML-based predictive modeling in food drying holds considerable potential for extracting drying kinetics information without resorting to computationally intensive physics-based numerical models. Additionally, ML predictive models (e.g., neural networks, decision trees, and SVM) play a crucial role in predicting food properties and microstructural features, eliminating the need for extensive experimental procedures [50, 51]. Determining food drying models relies heavily on understanding food properties, which are often challenging to obtain due to the complexities involved in experimental investigations. Furthermore, optimizing drying process parameters, such as drying time, temperature, air velocity, and humidity, is essential for energy efficiency and product quality. Challenges associated with applying ML-based approaches to multiscale modeling emphasized the need to advance this field by implementing ML-based strategies in multiscale modeling and properties prediction [46].

Existing research predominantly employs artificial neural networks (ANN) for food drying studies, but their complexity hinders process comprehensibility. The study suggests exploring alternative models such as decision tree (DT) or other regression-type models for improved comprehensibility. Additionally, hybrid ML-based algorithms, combining ANN, fuzzy logic, DT, or their combinations with classical mathematical models, are proposed to address challenges associated with single ANN applications. The insights presented in this paper are anticipated to contribute to the further advancement of ML-based approaches in food drying. Noteworthy applications of ML in drying include grapes and sugar beet pulp [52, 53].

Applications of NIRS, HSI, and CV with AI for Batch and Continuous Drying

NIRS, HSI, and CV enabled AI to streamline batch and continuous drying systems by providing real-time moisture analysis, optimizing drying parameters, and automating quality assessment processes. Batch drying involves drying a set amount of material at a time with manual parameter adjustments, while continuous drying maintains a constant flow of material with real-time parameter adaptation [54]. Continuous drying offers more uniform product quality and higher efficiency, particularly suitable for large-scale production, and both systems benefit from real-time data analysis [55].

Batch Process Ishikawa, Ueno, and Fujii [56] analyzed the water activity and moisture change of lump beef samples during batch drying with the integration of NIRS. The results demonstrated a linear decline in water activity assessed by NIR spectra, reaching 0.8 after roughly 22–25 days. Their study suggested that NIRS may be used to evaluate the dry status of beef at its surface without causing damage while it dries. The study effectively demonstrated surface water activity profiling using NIRS, indicating its potential for assessing beef drying conditions. Monitoring water activity is crucial for managing the quality of aged beef due to

surface bacteria. Previous research utilized NIR spectra (700–1050 nm) to estimate surface water activity, identifying optimal conditions for surface monitoring with samples of various thicknesses (2–10 mm). The moisture content prediction achieved R^2 of 0.81 and RMSE of 0.34 g/g-d.s. Achata et al. [57] studied NIR-HSI integrated beef jerky drying at 60 °C. Chemometrics and NIR-HSI estimated relative moisture content, yielding models with the ratio of performance to deviation (RPD) > 4 for process control, which indicates a very effective system for quality prediction. Top-performing models used EMCVS-selectivity ratio and ensemble Monte Carlo variable selection techniques, highlighting the potential of chemometrics and NIR-HSI as process analytical technology (PAT) tools for optimizing beef jerky drying. Amjad et al. [58] reported a smart CV system for large-scale drying to non-destructively monitor food quality changes (color dynamics, shrinkage, and weight loss) during convective drying. Potato slices tested the shiftable imaging system on a diagonal batch drier, validating its real-time quality monitoring in diverse drying conditions. The quadratic regression model accurately described correlations between product color, shrinkage, and moisture ratio, providing valuable insights for refining the drying process.

Continuous Process Leys et al. [54] established a novel methodology utilizing NIRS as an inline PAT tool to investigate desorption kinetics during freeze-drying. The focus was particularly on the secondary drying phase within an innovative spin freeze-drying technology. Notably, the product temperature during secondary drying exerted the most significant positive influence on the desorption rate. Enhanced drying rates were observed with higher cooling rates during freezing, while larger filling volumes tended to decrease the desorption rate. Interestingly, the 10 to 30 Pa chamber pressure showed no discernible impact on the desorption process. The findings suggest that inline NIRS has potential applications for studying desorption kinetics across various formulations, providing valuable insights into the underlying mechanisms. Moreover, this approach holds promise for determining optimal process settings to enhance desorption rates during the secondary drying stage of continuous freeze-drying processes (Fig. 12).

For the CV approach, empirical models offer a practical and straightforward approach for studying drying kinetics in continuous tray bed dryers, for achieving 12–13% moisture content at 60 °C took approximately 110 min. While mechanistic models provide a more comprehensive understanding, they demand advanced mathematical and computational skills. Drying models based on image data, mainly using convolutional neural networks, have high computational costs but are suitable for research applications. Simple classification algorithms utilizing RGB data from images effectively predict desirably and undesirably dried outcomes. Moreover, a smartphone-based tool emerged as a convenient and practical method for estimation, potentially finding applications in industries as a PAT tool [14].

Integrating advanced analytical tools such as NIRS, NIR-HSI, and RGB images has proven highly effective in batch and continuous drying processes. The successful application of NIR in batch drying facilitates real-time monitoring of water activity and moisture content in beef samples, offering valuable insights into the drying conditions. Similarly, in continuous freeze-drying processes, NIR serves as an inline PAT tool, providing a means to investigate desorption kinetics and optimize process settings. Moreover, the CV approach, encompassing HSI and RGB data, demonstrates its versatility in batch and continuous settings [14, 54, 56].

Advantages of AI-Enabled Optical Sensing for Drying Systems

The major key drawback of traditional drying process monitoring is the lack of real-time feedback, which can lead to over-drying or under-drying. This will result in inconsistent product quality and potential waste. Additionally, traditional methods often rely on manual sampling and analysis, which can be time-consuming, labor-intensive, and subject to human error. This will also impact the accuracy and reliability of the drying process control. AI-enabled optical sensing for drying systems offers numerous advantages that significantly enhance efficiency and effectiveness. Firstly, it provides real-time monitoring and control. So, it will allow for immediate adjustments to drying conditions, ensuring optimal drying and preventing over-drying or under-drying. This capability will lead to improved product quality and consistency. Secondly, AI algorithms can analyze vast amounts of data from optical sensors to detect patterns and predict outcomes. It will help proactive maintenance and reduce downtime by identifying potential issues before they become critical. Additionally, AI-driven systems can optimize energy usage by fine-tuning drying parameters. It will save energy and reduce operational costs. Moreover, these systems can be easily integrated with existing industrial setups with minimum modifications. Overall, AI-enabled optical sensing represents a transformative approach that drives efficiency, quality, and cost-effectiveness in drying processes [34].

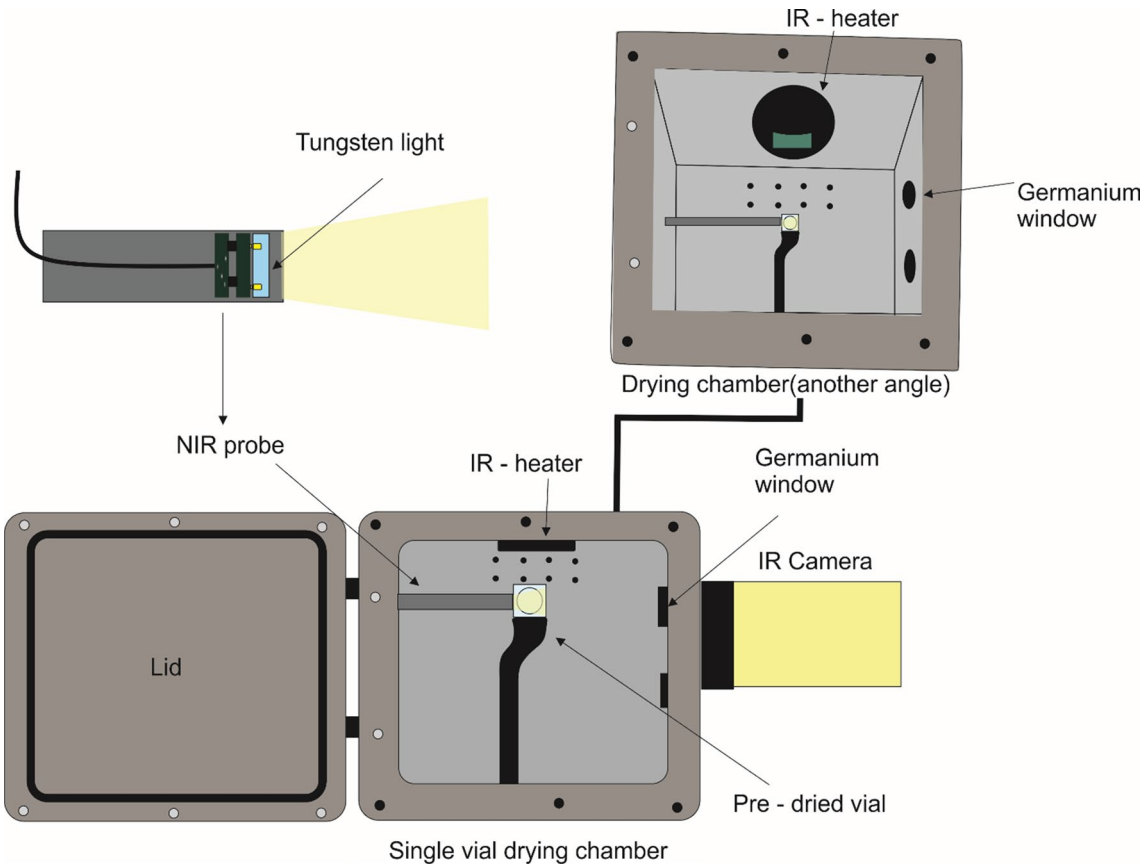


Fig. 12 A drying chamber designed for single-vial use and an experimental configuration for NIRS measurements, incorporating control over product temperature

Application and Techniques of AI-Enabled Optical Sensing Methods in Smart Drying

Near-Infrared Spectroscopy

NIRS is vital in monitoring the drying process as it provides detailed information on moisture content and chemical composition changes in real-time. This non-destructive technique enables precise and rapid assessment of drying efficiency to ensure product quality and consistency. The applications of NIRS with smart drying processes are presented in Table 1. Collell et al. [59] used NIR to monitor water activity and moisture content dynamics during the drying of fermented sausages. Sinelli et al. [60] used NIR to discern chemical, nutritional, and structural alterations in blueberries undergoing osmotic and air-drying processes, thereby augmenting the understanding of fruit preservation dynamics. Mensink et al. [61] applied NIR to freeze-drying methods and demonstrated its ability to evaluate hydrogen bond efficiency between protein and sugar. This not only improved freeze-drying techniques but also highlighted the ability of NIR to reveal key biochemical processes.

Campos et al. [62] expanded NIRS applications to predict real-time sodium content in dry-cured ham slices. This marked a significant step forward in quality control for the meat processing industry. Moscetti et al. [63] showed that NIR is effective in monitoring changes in organic apple slices during hot air drying for predicting water activity, moisture content, soluble solid content, and color changes. Phetpan, Udompetaikul, and Sirisomboon [64] further examined NIR's ability to measure moisture content during tapioca starch drying. Liu et al. [65] developed an intelligent method for detecting drying transitions and endpoints in blueberries during pulse-spouted microwave freeze-drying. They used a back-propagation ANN model to integrate high-frequency microwaves and advanced control systems for precise drying process management, as shown in Fig. 13. The configuration encompassed high-frequency microwaves at 2450 MHz, magnetrons strategically positioned around a cylindrical glass drying chamber, and an intricate control system facilitated by a data collecting module, a threshold analysis module, and an automatic control module, culminating in the precise modulation of spouting duration and spouting interval duration in the spouted bed.

Table 1 Application of NIRS in smart drying systems

Drying technique	Product	Parameters studied	Spectral range (nm)	Data analysis method	Drying conditions	Accuracy indicator(s)	Reference
Radio Frequency Convective Hot air drying	Apples Slices	Moisture Vitamin C	950–1650 Absorbance	[1]PCR PLSR BP-ANN	Dried at 60 °C 1.5 m/s 300 min	BP-ANN MC $R^2_p = 0.98$ $RMSE_P = 0.32$ (g/g) VCC, and $R^2_p = = 0.93$ $RMSE_P = 0.62$ (g/g)	[71]
Convective	Blueberries	Moisture Soluble solid content (SSC)	800–2780 Reflectance	PCA	Air drying 70ºC 8 h	-	[60]
Convective Hot air drying	Apple (var. Gala)	Moisture a_w SST	1100–2300 Reflectance	PLS PLS-DA	Hot air drying 60ºC, 8 h	aw $R^2_p = 0.97–0.98$ MC $R^2_p = 0.97–0.98$ SST $R^2_p = 0.96–0.97$ Chroma $R^2_p = 0.77–0.86$ $RMSE_p = 0.00–5.01$	[63]
Freeze Drying	Protein powder	Hydrogen bond formation	1000–2220 Reflectance	-	Dried at − 40 °C 0.5 °C/min 150 µbar 40 °C C in 350 min	NA	[61]
Convective hot air	Carrot/Apple	aw Moisture Carotenoids SST	1100–2300 Absorbance	PLSR	Dried at 50 °C/ 70 °C/8 h	All parameters $R^2 > 0.96$ $RMSE_P$ 0.03–0.04	[66]
Convective	Tapioca Starch	Moisture	950–1650 Reflectance	PLSR	Cyclone drying system. In-line NIR SNV	$R^2_p = 065$ $RMSE_P = 0.61\%$, Bias = 0.00%.	[64]
Microwave Freeze drying	Blueberry	Moisture	900–1599 Absorbance	[2]BP-ANN LF-NMR	Laboratory-scale microwave vacuum dryer 80 Pa 2450 MHz _80 ◦C for 10 h,	$R^2_c = 0.99$ $R^2_p = 0.98–0.99$	[65]
Freeze drying/conventional drying	Kakadu plum powder	Moisture	860–2530 Reflectance	2DCOS	Dried at − 48 °C for 7 days / C: 45 °C for 3 to 4 days	-	[67]
Ultrasonic drying	Apple slices	Moisture endpoint	900–1700 Absorbance	MLR PLSR	Vibration 20khz	$R^2_p = 0.97–0.99$ $RMSE_p = 1.51–2.95\%$	[68]
Ultrasonic Hot air drying	Apple slices	Moisture endpoint	900–1700 Reflectance	PCA PLSR	Dried at 60°C 1 m/s 90–115 min	$R^2_p = 0.91–0.95$ $RMSE_p = 0.29–0.81\%$	[69]
Hot air drying	Apple slices	Moisture endpoint	1550–1950 Absorbance	PLSR ANN	Lab scale Smart Dryer 50 °C, 1.0 m/s	$R^2_p = 0.91–0.93$ $RMSE_p =$ 0.55–0.59%	[70]

[1]**PCR**: Principal Component Regression, [2]**BP-ANN**: Backpropagation Artificial Neural Network, **aw**: Water Activity, [3]**PLC**: Programmable Logic Controller

Sturm et al. [66] studied NIRS to monitor real-time changes in apples and carrots. The single-point NIRS assessment indicated the potential for proactive and non-destructive monitoring of quality parameters like moisture content, water activity, carotenoids, color, and soluble solid content. Cozzolino et al. [67] investigated the impact of drying methodologies on Kakadu plum powder through the fusion of two-dimensional correlation spectroscopy (2DCOS) and NIRS. Recent studies by Malvandi, Feng, and Kamruzzaman [68] explored miniature NIRS with multivariate analysis for real-time moisture content evaluation in apple slices during direct-contact ultrasonic drying. They developed a precise MLR model using three specific wavelengths (1359, 1517, and 1594 nm) and achieved high accuracy expressed by Predicted R^2 and Root Mean Square Error of Prediction ($R^2_p = 0.99$, $RMSE_P = 3.32\%$). The study highlights the effectiveness of low-cost NIRS for real-time, non-destructive evaluations in industrial drying processes. Kapoor et al. [69] investigated the use of miniature NIRS for fast monitoring and predictive modeling of the drying

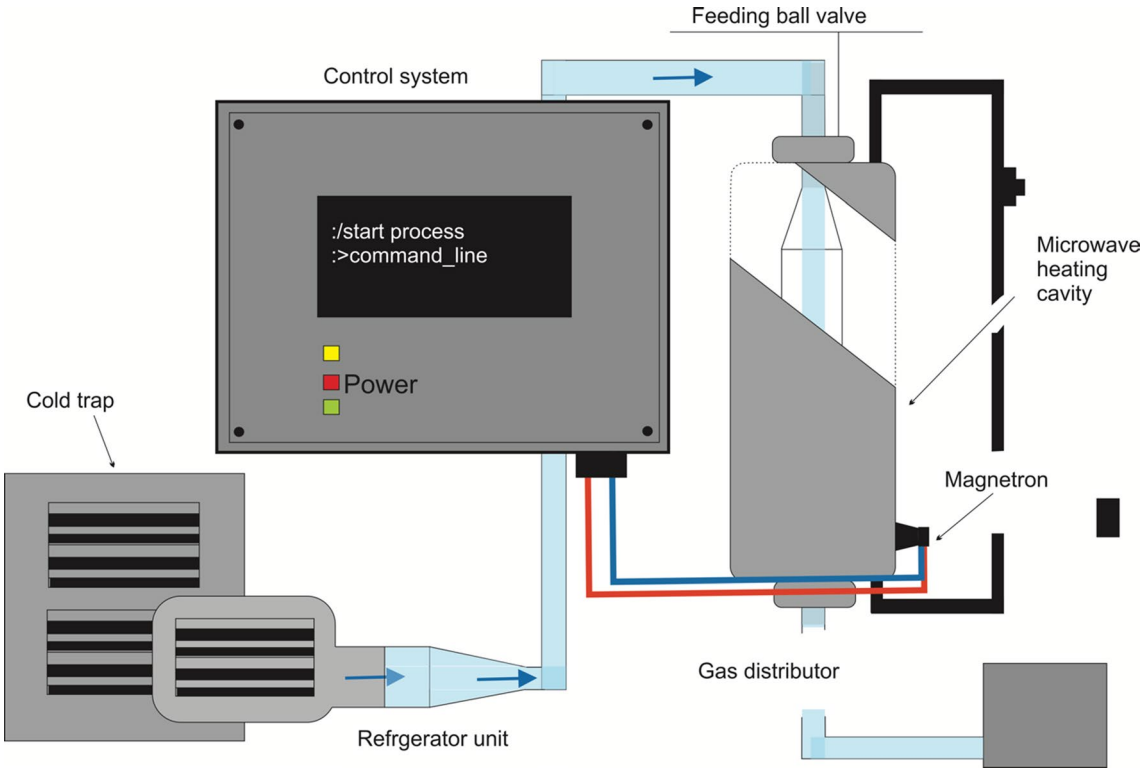


Fig. 13 Smart dryer LF-NMR and NIR in pulse-spouted microwave freeze drying

process of edible coatings on apple slices. They used PLSR on raw and pre-treated spectra and identified seven critical wavelengths for easy monitoring of moisture content. The study confirmed that miniature NIRS can effectively monitor the drying process, distinguish between coated and uncoated slices, and detect differences in drying times based on sugar and water absorption characteristics. They suggested creating a robust system for real-time monitoring with three key wavelengths and encouraging future research on using low-cost spectral sensors for predictive modeling of apple moisture content during drying.

Significant progress has been made in designing smart drying systems incorporating digital cameras and NIR sensors. For example, in collaboration with the Center for Advanced Research in Drying (CARD, NSF-Industry initiative), the Illinois Smart Analytical Lab (ISAL) developed an intelligent dryer using advanced technology and the Internet of Things (IoT) sensors (Fig. 14). This smart dryer includes a digital camera, an NIR sensor, and traditional sensors for measuring temperature, humidity, and air velocity. It can be used to dry fruits and vegetables, monitor the process in real time, and automatically determine the drying endpoint [70].

Recently Jin et al. [71] combined a hot-air assisted radio frequency drying (HARDFD) (Fig. 15) with NIR to assess moisture content (MC) and vitamin C content (VCC) of apple slices. Statistical models, including principal component regression (PCR), partial least squares regression (PLSR), and back propagation-artificial neural networks (BP-ANN), were developed. The BP-ANN model showed the highest prediction accuracy, with $RMSE_P$ values of 0.33 for MC and 0.61 for VCC, and R_p^2 of 0.98 and 0.93, respectively. This study highlights the effectiveness of NIR spectroscopy combined with BP-ANN for rapid, non-destructive quality assessment during HARFD.

Hyperspectral Imaging

HSI is crucial in monitoring the drying process as it captures detailed spectral and spatial information across a wide range of wavelengths. This technology allows for precise and non-destructive assessment of moisture content, texture, and chemical composition changes in real-time. By providing comprehensive data, HSI helps optimize drying parameters, ensuring product quality and consistency while reducing energy consumption and processing time. Its ability to detect subtle variations and predict drying endpoints makes HSI an invaluable tool in improving efficiency and accuracy in various drying applications. Many studies have reported HSI as a rapid and robust method for instantaneous moisture determination using different algorithms (Table 2). Recently, shortwave infrared HSI was employed for moisture content detection of persimmons during convective drying [10]. They constructed calibration models using standard normal variate (SNV), multiple scatter

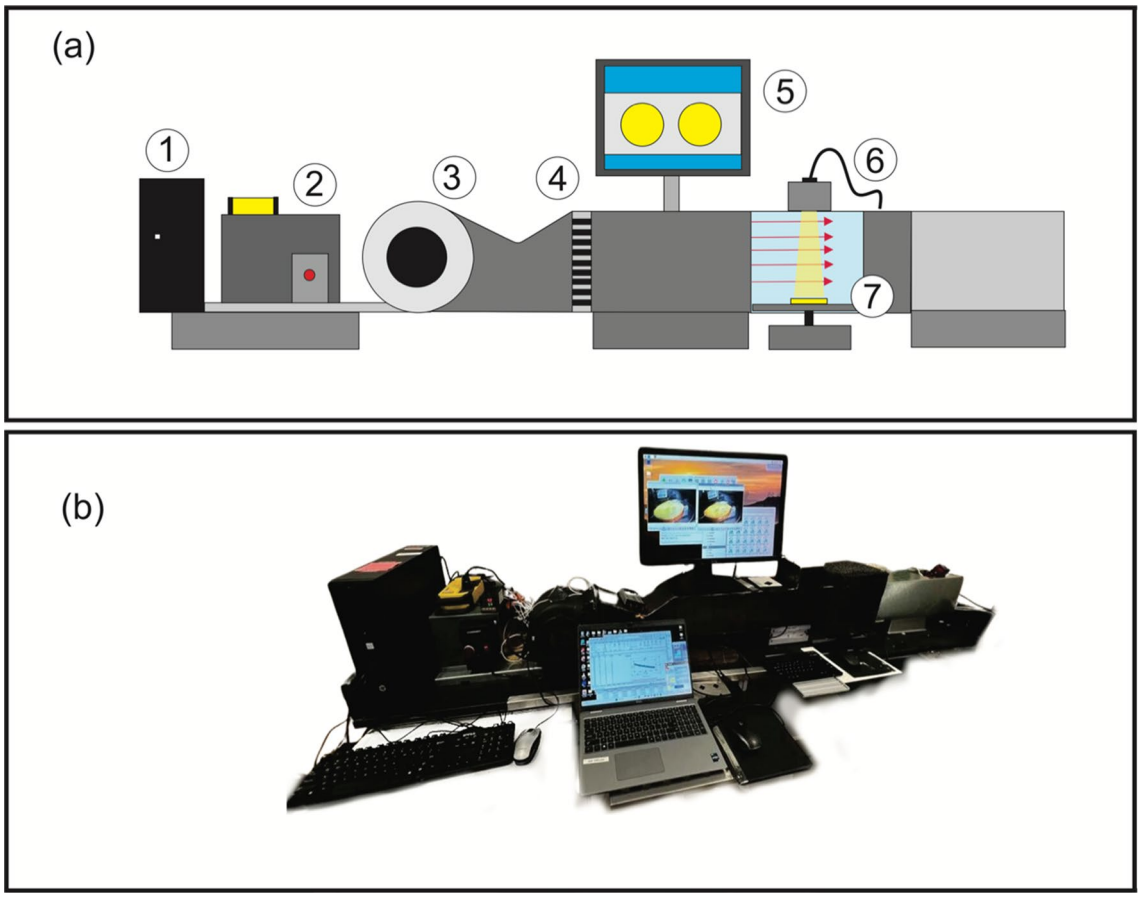


Fig. 14 Smart drying system developed in Illinois Smart Analytical Lab (ISAL) with compact NIRS and RGB imaging sensors; (**a**) (**1**) Computer; (**2**) Power system; (**3**) Fan; (**4**) Heater element; (**5**) Computer screen; (**6**) RGB and NIR sensors; (**7**) Sample chamber. (**b**) Real equipment [70]

correction (MSC), normalization, and Savitzky-Golay (SG) pre-processing followed by PLSR. The best calibration model had an R^2 value of 0.97, with 95% model precision. Results suggested that HSI technology can detect the moisture content of fruits during convective drying. For moisture prediction during soybean drying, HSI in the spectral band 400–1000 nm was reported by [72]. Using a binary firework algorithm (BFWA), they selected 12 important wavelengths for short-form PLSR model development. The PLSR model with BFWA achieved a prediction accuracy of R_p = 0.97% and $RMSE_P$ = 5.11%, indicating the system's effectiveness in predicting moisture during drying. Vis-NIR HSI has been evaluated for moisture prediction of mango slices during microwave-vacuum drying [73]. They followed three waveband selection strategies for accurate moisture prediction. R^2_p = 0.97% and $RMSE_P$ = 4.61% have been reported for the top model. In addition to chromaticity, HSI has reported the moisture content of potato slices during convection drying in the visible NIR region [74]. The PLS model with R^2 = 0.93–0.98 and RMSE = 0.16–0.36 at six optimal wavelengths showed an ability of HSI for moisture prediction. Ren and Sun [75] used HSI system for beef moisture prediction during microwave and hot-air drying and achieved high prediction accuracy (R^2_p = 0.99 for moisture and R^2_p = 0.99 for drying rate). HSI in the 951–1630 nm range has successfully measured moisture and freezable water during microwave-vacuum drying [73]. Using only seven features, a short-form PLSR model achieved excellent precision (R^2_p = 0.99, $RMSE_P$ = 1.28%). Yu et al. [72] developed HSI system (400–1000 nm) for microwave-assisted pulse-spouted bed vacuum-drying of vegetable soybeans and accomplished good performance (R^2_p = 0.90, $RMSE_P$ = 9.2%).

The color of food products is an important quality attribute that can influence consumer acceptance, and accurate measurement is essential for maintaining consistent product quality. HSI offers a non-destructive, non-contact method for measuring food color and chromaticity, providing detailed spectral information that can be used for quality control and product development in the food industry. To measure food color using HSI, a hyperspectral camera captures images of the food product at multiple wavelengths across the visible and NIR spectrum. The resulting image contains a spectrum for each pixel, allowing for the analysis

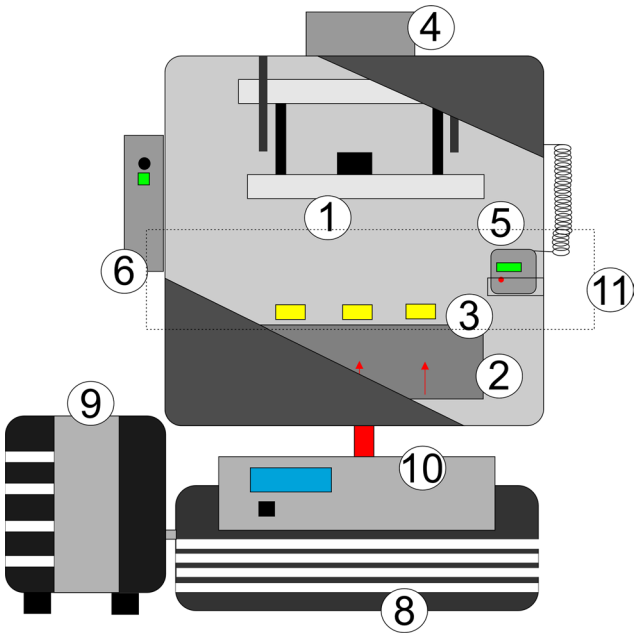


Fig. 15 Hot air-assisted radio frequency (RF) drying system: (**1**) upper electrode, (**2**) lower electrode, (**3**) apple slices, (**4**) exhaust air fan, (**5**) electric controller of RF heater, (**6**) safety switches, (**7**) hot air tube and distributors, (**8**) electric heater, (**9**) fresh air fan (**10**) electric controller of hot air system, (**11**) conveyor belt

of the spectral signature of the food product. Chemometric techniques, such as principal component analysis or partial least squares regression, are then used to the spectral data to identify the color attributes of the food product. These approaches may also be utilized to construct prediction models that can be used in real-time to determine color attributes, allowing for more efficient quality control. Saleh et al. [76] developed an excellent prediction color model of carrot using Vis-NIR HSI technology in a hot air-drying system. Very good model performance was found for total carotenoids ($R^2_p = 0.64$, $RMSE_p = 32.62$), L* ($R^2v = 0.68$, $RMSE_p = 32.62$), a* ($R^2_p = 0.69$, $RMSE_p = 1.18$), and b* ($R^2_p = 0.60$, $RMS_p = 1.45$). They used five identical features to develop a robust prediction model based on VIP scores. Color prediction of purple-speckled cocoyam slices during hot-air drying has been studied using HSI in the spectral range of 400–1700 nm [77]. Using the partial least squares beta coefficients (PLS-BETA) and variable importance in projection (PLS-VIP) techniques, they included 19 significant wavelengths into an effective prediction model with a satisfactory result for color ($RMSE_P < 8.0$, $R^2_p = 0.70$–0.90, $RPD > 2.0$). Amjad et al. [74] published an HSI method for predicting the chromaticity of sliced potatoes under convective drying. PLS models for CIELAB a* performed well ($R^2 = 0.91$ to 0.65, $RMSE = 0.61$–1.61) in predicting chromaticity by employing only six wavelengths. Crichton et al. [78] published a similar investigation on the chromaticity of pre-treated apple slices, using the same few wavelengths to predict CIELAB a* (543, 966 nm) and CIELAB b* (510, 664, 714, 914, 969 nm). The feasibility of HSI in conjunction with chemometrics analysis for the quick and non-destructive measurement of anthocyanin concentration in purple-fleshed sweet potatoes during ultrasound-assisted hot air drying has been investigated by Liu et al. [79] Three techniques, including PLSR, LS-SVM, and MLR, were utilized to construct models based on ten preferred wavelengths chosen using the regression coefficients approach to simplify the calibration model. The model's performance ($R^2_c = 0.87$, $R^2_p = 0.87$) revealed that the HSI approach is effective for quick and non-destructive assessment of the anthocyanin concentration.

HSI can be used to determine the textural properties of a material or surface by analyzing the spectral information of the image. Texture analysis characterizes the spatial arrangement of the pixels or features in an image. HSI technology can also be combined with ML algorithms to automatically classify and identify different textural properties of food for online inspection and quality control. HSI system is successfully used for hardness and shrinkage determination during microwave-vacuum drying [80]. A PLSR model based on optimum wavelengths demonstrated the efficiency of the HSI system with an R^2_p of 0.97 and an RMSEP of 4.63. Soluble solid is an important parameter affecting food's taste, texture, shelf life, and nutritional properties. Chen et al. [81] proposed a technique based on HSI to undertake online and non-destructive assaying of soluble solids and water-soluble tannin in persimmon fruits during drying. PCR, PLSR, LS-SVR, and radial basis neural network (RBFNN) were used to construct a predictive model. These findings suggest that HSI effectively determines a product's soluble solids and chemical contents during drying. An interesting study on the online monitoring of meat's total viable basic nitrogen (TVB-N) content during convective drying was performed in the spectral range of 400–1000 nm [82]. Multivariate calibration models were constructed using PLSR, MLR, and LS-SVM approaches. The applicability of multispectral imaging methods for signal acquisition of the TVB-N content of foods throughout drying processes was shown by using nine key wavelengths to develop a prediction model.

Arefi et al. [83] studied the combination of drying technology and digital twins, suggesting affordable imaging alternatives to hyperspectral sensors (Fig. 16). Using 980 nm LEDs and band-pass filters, they accurately measured moisture content, soluble solids content (SSC), and shrinkage during apple slice drying at 60 °C. Gaussian process regression models showed high precision, with moisture content predictions achieving R^2 values of 1.00 and 0.99 for LED and BPF, respectively. These models, applicable to external apples, confirmed their robustness. Integrating 980 nm LEDs into an experimental dryer allowed real-time

Table 2 Application of hyperspectral imaging technology in smart drying systems

Drying technique	Product	Parameters studied	Spectral range (nm)	Data analysis method	Drying conditions	Accuracy indicators	Reference
Spray Dryer	Tomato Powder	Material loss	NA	Neuro -fuzzy	Tomato paste (30% dry matter), hot break technology (85–90ºC), final moisture content 47–53%	-	[84]
Convective	Potato	Moisture, Chromaticity	500–1000 Reflectance	PLSR	5 mm, 7 mm, 9 mm slices; 50ºC, 60ºC, 70ºC; 12% moisture	Moisture: R_p^2=0.93–0.98, $RMSE_p$=0.16–0.36% Chromaticity: R_p^2=0.91 − 0.65, $RMSE_p$=0.61–1.78%	[74]
Convective	Persimmon fruits	Water soluble tannin, Soluble solids	470–900	PLSR, PCR, LS-SVR, RBFNN,	Drying duration: 12 h to 96 h; Sharp drop in tannins from 60 h to 96 h	R^2_c = 0.74 $RMSE_P$=1.75 R^2_p = 0.93 $REMSE_c$=0.99	[81]
Convective	Persimmon fruits	Moisture	900–2000	PLSR	12 h drying, 12 h rest/day; 1 to 6 days; 20 samples per group	R^2_c = 0.97 RMSEc = 1.61 R^2_p = 0.94 $RMSE_p$ =2.13%	[10]
Microwave-vacuum	Ginger slice	Hardness, shrinkage,	950-1655 Reflectance	PLSR	Different geometries: original, cylinder, cuboid; Drying time: 80 min; Significant differences ($P < 0.05$) in shrinkage rates	R^2_p = 0.97 $RMSE_P$ =4.63%	[81]
Ultrasound-assisted Convective	Purple-fleshed sweet potato	anthocyanin content	371–1023 Reflectance	PLSR, LS-SVM, MLR	Dried at 40 °C, 1 m/s; 60 W, 28 kHz ultrasound; 0 to 7 h drying periods; 132 slices total	R^2_c = 0.86, R^2_p = 0.86 $RMSE_P$ =0.632 (mg/g)	[80]
Convective	Potato Tubers	Moisture content	733–970 Reflectance	PLSR	Drying at 60 °C for 2 days; Moisture content: 50–80%	R^2_p = 0.53 RPD = 1.46 $RMSE_p$=5.04%	[104]

Table 2 (continued)

Drying technique	Product	Parameters studied	Spectral range (nm)	Data analysis method	Drying conditions	Accuracy indicators	Reference
Convective	Purple-speckled cocoyam	Moisture, color, chemical and structural attributes	400–1700 Reflectance	PLS-BETA, PLS-VIP	Drying temperature: 40, 60, 80 °C; Air velocity: 2.0 ± 0.2, 3.0 ± 0.4, 4.6 ± 0.7 m/s; Preheated for 30 min; Final moisture ~ 0.11 kgW/kgDM; Samples drawn every 30 min	Moisture $R^2_c = 0.99$ $R^2_p = 0.99$ $RMSE_c = 1.70$ $RMSE_P = 1.98$ Color Attribute $R^2_c = 0.99$ $R^2_p = 0.99$ $RMSE_c = 7.10^0$ $RMSE_P = 7.60^0$ Structural Attribute $R^2_c = 0.97$ $R^2_p = 0.96$ $RMSE_c = 0.042$ $RMSE_P = 0.039$ TPC $R^2_c = 0.77$ $R^2_p = 0.76$ $RMSE_c = 0.063$ µg/g $RMSE_P = 2.2$ µg/g	[77]
Pulse spray Microwave- vacuum	Soybean	Moisture	400–1000 Reflectance	PLSR (BFWA)	The drying was performed using pulse spray microwave vacuum drying temperature 0–80 min	$R_p > 0.93$ $R_c > 0.94$ $RMSE_P = 4.38$-6.71% $RMSE_c = 5.10$-7.41%	[72]
Microwave-vacuum	Mango slice	Moisture	400–1000 Reflectance	PLSR	Drying temperature: 105 °C; Drying duration: 24 h; Relative humidity controlled	$R^2_c = 0.98$ $R^2_p = 0.99$ $RMSE_c = 3.53\%$ $RMSE_p = 4.48\%$	[105]
Microwave-vacuum	Mango slice	Moisture	951–1630 Reflectance	PLSR	Heating; drying times: 0–30 min; MC by oven-drying (104 °C, 24 h)	$R^2_c = 0.99$ $R^2_p = 0.99$ $RMSE_c = 1.36\%$ $RMSE_p = 1.12\%$	[73]
Microwave-vacuum	Beef slices	Moisture, rehydration rate	950–1654 Reflectance	RC-SG-PLSR	Frozen at -18 °C; Sliced to 5 mm and 40 mm × 40 mm pieces; Total 216 samples; Thawed at 4 °C in sealed bags	$R^2_c = 0.92$ $R^2_p = 0.91$ $RMSE_c = 2.86\%$ $RMSE_p = 3.31\%$	[75]

Table 2 (continued)

Drying technique	Product	Parameters studied	Spectral range (nm)	Data analysis method	Drying conditions	Accuracy indicators	Reference
Convective	Carrot slices	carotenoids, color, and moisture	400–1010 Reflectance	PLSR	HT mini dryer; 0.6 m/s; 200 g at 50–70 °C; Initial moisture ~ 88%; Three replicates; Every 30 min; Final moisture ~ 10%	Moisture R^2_c = 0.92 R^2_p = 0.91 $RMSE_c$=8.15% $RMSE_p$=8.16% Color R^2_c = 0.67–0.78 R^2_p = 0.60–0.69 $RMSE_c$=1.06–156% $RMSE_p$=1.18–1.75%	[76]
Convective	Shrimp	moisture, color, and texture	400–1000 Reflectance	PLSR, LS-SVM	Dried at 55 °C for 12 h; Samples at 1–12 h (16 samples/time point, 104 total)	Moisture R^2_p = 0.69-73 $RMSE_p$=8.56–9.56% RPD = 1.06–1.19 Texture *Hardness* R^2_p = 0.69 $RMSE_p$=46.84 N RPD = 1.39 *Elasticity* R^2_p = 0.70 $RMSE_p$=0.20 mm RPD = 1.39	[106]
Convective	Cured meat	TVB-N	400–1000 Reflectance	PLSR, LS-SVM, MLR	Dried 48 h (58 °C to 52 °C); 15 samples (0–48 h); 210 slices (25 g) at 4 °C	R^2_c = 0.97 $RMSE_c$=2.08 R^2_p = 0.86 $RMSE_P$=4.73	[82]
Convective	Apple	Moisture, Total solid	400–1750 Reflectance	GPR	Dried at 60 °C for 195 min (~ 14% moisture); 224 slices (7 replications); 252 total; Stored at -70 °C	Moisture R^2_c = 0.986 R^2_p = 0.99 $RMSE_c$=2.93% $RMSE_p$=2.45% SSC R^2_c = 0.88 R^2_p = 0.87 $RMSE_c$=7.24% $RMSE_p$=7.485% Shrinkage R^2_c = 0.98 R^2_p = 0.98 $RMSE_c$=3.89% $RMSE_p$=4.29%	[83]

GPR: Gaussian Process Regression, **PLSR**: partial least squares regression, **PCR**: principal component regression, **LS-SVR**: least squares support vector regression, **RBFNN**: radial basis function neural network, **VCPA**: variable combination population analysis, **BFWA**: binary firework algorithm, **BPN**: backpropagation network

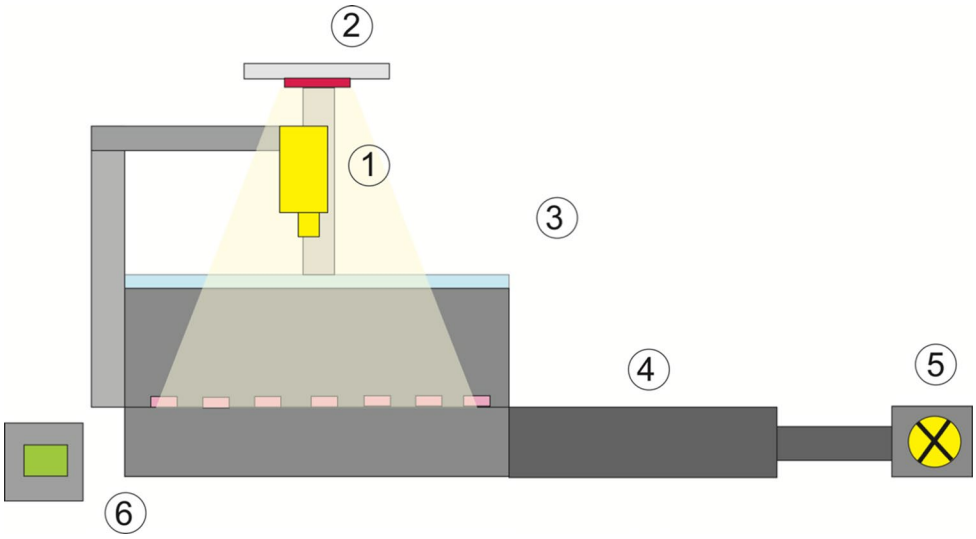


Fig. 16 Inline monitoring setup: (**1**) Camera, (**2**) 980 nm LED arrays, (**3**) drying chamber, (**4**) heater, (**5**) air pump, and (**6**) control unit

temperature control based on moisture content, advancing affordable and simple imaging sensors for smart dryers using digital twins. Zubkova and Sherstjuk [84] created an intelligent control system using HSI for a spray drying machine to produce tomato powder. Due to the complexity of the process, traditional mathematical methods were not feasible. Their solution involved two stages: perception and decision-making. In the perception stage, electro-optical and infrared sensors captured images, and a four-layer neural network identified the mixture's state. This integration allowed deviations in the process flow to be detected. In the decision-making stage, a neuro-fuzzy controller based on the Takagi-Sugeno model was implemented using the adaptive neuro-fuzzy inference system (ANFIS) model. This system ensured reliable and efficient control of the spray drying process, reducing errors, raw material losses, and the overall cost of the final product.

Computer Vision

Drying causes significant changes in the physic-chemical and organoleptic properties of food materials. Due to the complexity and nature of the drying process, mechanical changes like non-homogeneous shrinkage, shape distortion, casehardening, glass transition, and surface cracking are hard to measure online or in-line during drying. Also, heat-sensitive components such as vitamins, pigments, and flavor elements are almost impossible to evaluate on the processing line during the drying [85] and are traditionally assessed at the end of the drying process. Many studies reveal that CV can effectively optimize and control industrial drying processes, which may lead to significant economic benefits and boost productivity. Recent CV applications for food drying are listed in Table 3.

Color is a critical food property that significantly influences consumers' perception of food quality, freshness, and taste. Plant and animal tissues such as fruits, vegetables, and meat exhibit a change in color during drying, which is affected by the temperature, air relative humidity, pretreatments, and drying methods. Color measurement is traditionally carried out using colorimeters, whose viewing area usually is 2–5 cm^2; thus, it is inappropriate to discriminate the overall color change of the whole dried food. The spectrophotometric evaluation of the whole product is time-consuming, requires sample preparation, and is unsuitable for rapid analysis of the color changes during drying [1, 86]. The combination of CV and ML methodologies exhibits profound capabilities in precisely quantifying color attributes throughout the complex drying process. Encompassing crucial parameters such as hue, saturation, brightness, and overall color variation over temporal intervals, this technological synergy has gained considerable attention in recent scientific research.

Chakravartula et al. [42] found a clear correlation between brightness (L*) and color (hue) parameters with the water content in carrot samples. Similarly, Moscetti et al. [87] studied the drying dynamics of carrot slices at 35 °C for 36 h using a smart dryer system with a digital camera. Their linear models showed excellent predictive potential with an R^2 over 0.99.

Hosseinpour et al. [88] used an online CCD camera to study the effect of temperature and air velocity on color changes during drying, measuring parameters like brightness, redness, yellowness, total color difference, chroma, hue angle, and browning index with high accuracy (low RMSE values). Onwude et al. [89] explored laser-induced back-scattering imaging with CV for sweet potato drying and found a strong link between temperature, moisture content, and color attributes, with R^2 over 0.70. Raponi et al.

Table 3 Application of computer vision in smart drying systems

Drying technique	Product	Parameters investigated	System components/Drying conditions	Accuracy indicator(s)	Reference
Convective	Carrot	Weight, color, and size	**CV components**: CMOS camera (DFK 33UX264) with LED illumination (4200 K), DC 24 V power supply **Drying conditions**: 35 °C, 35% RH, 3 m/s airflow, 36 h drying of 350 g carrot slices, real-time data.	Adjusted R^2: approx. 0.99, Reduced χ^2: approaching 0, RMSE (CNT): 0.007–0.174, RMSE (BL): 0.005–0.115	[42]
Convective	Carrot	Weight, size, and color	**CV components**: CMOS camera (Model: DFK 33UX264, The Imaging Source, Germany) with LED illumination (4200 K), DC 24 V power supply	Adjusted $R^2 >$ 0.99, RMSE similar to thin-layer models, MBE and reduced χ^2 tending towards zero	[87]
Spray drying	Rhubarb Juice Powders	Color features	**Drying conditions**: 35 °C temperature, 35% relative humidity, 3 m/s airflow, 36 h **CV components**: Nikon D5100 camera (16.2 MP, DX), Nikkor 18–105 mm lens, CFM DT-1308 luxmeter. **Drying Conditions**: 30–50% dry juice, DE 12–26, moisture 1.95–2.80%, water activity 0.14–0.21.	PCA correlation Adj.$R^2 > 0.97$ ANN classification $R^2_c = 0.91$ $R^2_p = 0.91$ RMSE ~ 0.04	[105]
Convective and steam	Shrimp	Color	**CV components**: HMI (FATEK FV035ST-C10), two digital cameras (Canon G9), MATLAB 7.7 for data logging. **Drying conditions**: HAD temperatures (50, 70, 90 °C), SSD temperatures (110, 120 °C), 3 air velocities (1, 1.5, 2 m/s), 6500 K fluorescent lamps, camera distance 40 cm, shutter speed 1/8s, max resolution 4000 × 3000 px.	For all models Adjusted R^2 Range: 0.3917–0.9992 RMSE Range: 0.0017–8.7626	[88]
Convective and microwave	Tarhana (Fermented wheat flour	Quality based on drying conditions	**CV components**: Canon EOS 500D (18–55 mm lens, 4,752 × 3,168 px), 2 × 23-W cool white fluorescent bulbs (25 cm above sample), Petri plate, dark room setup. **Drying conditions**: SD: 5 days at 20 °C, OD: 48 h at 55 °C, MD: 15 min at 900 W, 2,450 MHz. Moisture content monitored with OHAUS MB45 analyzer to 10%.	SDA correlation Adj. R^2: > 0.99 K-NN classification Accuracy: 99.5% ANN classification Accuracy: 98.6% RMSE ~ 0.04	[106]
Convective	Mushroom	Surface wrinkling and shrinkage	**CV components**: Basler Aca2500-14GC industrial camera (GigE), Computer M1214 lens, 6500 K LED (9 W), camera/light cooling system (air/water-cooled). **Drying conditions**: 60 °C, airflow 3.2–4.2 m/s, 4 RH control methods (40%, 30%, 20%, step-down), no external humidifier.	-	[97]
Convective	Corn	Impurity	**CV components**: The CMOS sensor (OV7725) by Howell Technology Co., Ltd. is a low-power, cost-effective device with strong performance in low-light and varying temperatures. It captures 8-bit JPG RGB images using a 3.6 mm lens with a 78° field of view. **Drying conditions**: Initial moisture content 27.9%, drying time 9 h, 5 images taken every 20 min during drying process.	F1 of broken corncob recognition: 83.05% F1 of broken bracts recognition: 83.87% F1 of crushed stone recognition: 87.43%	[102]
Hybrid (Convective-Infrared)	Kiwifruit	Shrinkage	**CV components**: Digital camera (COOLPIX P510, Nikon Co., Japan), MATLAB, genetic algorithm **Drying Conditions**: Drying experiments were conducted at air temperatures of 50 °C, 60 °C, and 70 °C, with air velocities of 0.5 m/s, 1 m/s, and 1.5 m/s.	-	[95]

Table 3 (continued)

Drying technique	Product	Parameters investigated	System components/Drying conditions	Accuracy indicator(s)	Reference
Convective	Sweet potato	Shrinkage	**CV components**: RGB image capture at 1 h intervals, pre-processing & segmentation (MATLAB), thresholding, erosion & dilation for image parameter extraction. **Drying conditions**: 50, 60, 70 °C, 1.2 m/s air velocity, RH 45–60%, moisture content expressed as moisture ratio (MR).	Moisture Content: R_c^2 = 0.772, $RMSE_c$ = 0.154; R_p^2 = 0.718, $RMSE_p$ = 0.175/ Color R_c^2 Range: 0.540–0.803, $RMSE_c$ Range: 0.575–3.859; R_p^2 Range: 0.462–0.758, $RMSE_p$ Range: 3.281–6.545	[90]
Convective	Sweet potato	Moisture content and color changes	**CV components**: CCD camera (QICAM Color Fast 1394), zoom lens (F5.6, 18 mm), laser diode (658 nm), Image-Pro Insight 9 software; fluorescent lamps (HFB RB 218). Covered chamber. **Drying conditions**: Convective hot-air dryer; 35 sweet potato samples, sliced; temperatures of 50, 60, and 70 °C; air velocity 1.2 m/s; thicknesses 2, 4, and 6 mm; RH 35–50%; slices removed hourly.	For color change: $R^2 > 0.7$ For moisture: $RMSE_p$ = 0.18	[96]
Far-infrared	Ginger	Drying optimization	**CV components**: Illumination system with rectangular LED lamp (25 cm × 4.6 cm, 6500 K) placed outside the door glass, CCD camera (MER-230, Daheng Imaging Co., Ltd, China), integrated computer hardware and software. **Drying conditions**: Far-infrared oven (3200 W) with remolded circuit for power control. Constant temperature drying at 50, 60, 70, and 80 °C.	-	[103]
Convective	Gala apple	Shape and size, moisture content, and color	**CV Components**: CMOS camera (EOS 400D, Canon, Japan) , Illumination source , DC 12 V circuit breaker with relay module (SRD T73, Ningbo Single Relay, China) ,Raspberry Pi B+ (Raspberry Pi Foundation, UK) -,300-W DC power supply (300XX, Codegen Technology, China) **Dryer conditions**: Biosec domus B5 (Tauro Essicatori srl, Italy). drying chamber: 26.50 L (~240 mm × 240 mm × 460 mm) Drying temperature: 60 °C (based on pre-screening tests; optimal for colorimetric parameters)	|BIAS| < 0.0085 RMSE < 0.04, Adj-R^2 = 99%	[43]
Convective	Apple slice	Area, thickness, color	**CV Components**: LED lighting (6000–6500 K, 400 lx) is positioned above, with a top camera (Unibrain Fire-I) 20 cm above the tray and a side camera (DFK31AF03) at 90° for side views. **Drying conditions**: Tray dryer (UOP8, Armfield Ltd.) modified for 40–80 °C, air velocity 0.9 m/s. For each experiment, the dryer was primed at 40, 60, and 80 °C.	Correlation $R^2 > 0.9$ Regression R^2 = 0.945, $p < 0.05$)	[92]
Convective air	Apple And Carrots	Physical Chemical Changes	**CV Components**: Specim V10E PFD hyperspectral camera (Schneider Optics Xenoplan 1.9/35), linear translation stage, and three 60 W halogen GU10 bulbs for illumination. **Drying System**: Apples sliced to 5 mm thickness, dried in a hot air convection oven at 50 °C. Convective air cabinet dryer (HT-Mini) at 50 and 70 °C.	Carrots SST $R_p^2 \geq 0.80$ Aw $R_p^2 \geq 0.80$ $RMSE_p$ = 0.03–0.04 Apples All Parameters $R_p^2 \geq 0.96$ $RMSE_p$ = 0.04–0.05	[66]

Table 3 (continued)

Drying technique	Product	Parameters investigated	System components/Drying conditions	Accuracy indicator(s)	Reference
Convective air	Apple	Color	**CV Components**: UI-5240CP-C-HQ camera (iDS Imaging) with built-in heater, LED bar light (6500 K, Stemmer Imaging). Mirror at 45° for side deformation. **Drying Conditions**: 3 m x 1.5 m x 1 m, 50 m² tray area, electric heater, ventilator, heat pump for RH. 40 °C drying temperature as reference case.	NA	[86]
Convective (Solar)	Turmeric	Color, morphology, texture, browning factor, and shrinkage	**CV Components**: Sony DSC WX350 camera (18.2 MP), LED strip, and two 7 W bulbs. Backlight is used for morphology, while front light (LED strips at 45°) is for texture. **Drying Conditions**: 70 g samples in a solar dryer, insulated with mineral wool, black interior, 3 mm glass cover, wire mesh tray, and air vents for circulation	NA	[97]
Convective	Papaya	Color	**CV Components**: Lightproof chamber with two 6500 K fluorescent lamps; Nikon COOLPIX P300 camera (12.5 cm from sample, 1/8 s shutter speed, F3.3 aperture). Six faces captured. **Drying conditions**: High-precision lab dryer (University of Hohenheim), 72 samples at 50 °C, 60 °C, 70 °C, and 80 °C, 25 g/kg dry air, 0.5 m/s airflow. Final moisture content: 0.16–0.18 kg/kg, water activity 0.5 ± 0.05.	$R_c^2 > 0.9$ $R_V^2 > 0.88$ MAPE = 10.75, 12.25	[13]
Fluidized bed	Peas	Shrinkage	**CV Components**: Nikon D5300 camera on 15 mm LWS Rail, Infinix Note 4 Pro for offline images, 2 R300 SEVR LED ring lamps (5,600 K, 1,187 lx). **Drying Conditions**: 350 g peas dried from 76–24% (wb) at 50 °C, 55 °C, 60 °C. 10 images every 15 min (2,992 × 2,000 px).	NA	[94]
Convective	Shrimp	Texture kinetics	**CV Components**: Two color cameras (Canon G9 digital color camera, Japan, and SDN-550P, day and night, Samsung, Korea), Zoom-Browser software. **Drying Conditions**: temperatures of 50–90 °C and superheated steam drying (SSD) temperatures of 110–120 °C with drying medium velocities of 1–2 m/s	High variability depending on the drying kinetics applies $R^2 = 0.00–0.99$ RMSE 0.210^{-5} − 9.24	[100]

MAPE: mean absolute percentage error, BIAS: A measure of the systematic error or the average difference between predicted and actual values. Lower values indicate less bias in the model. Reduced χ^2 (Chi-squared): A goodness-of-fit measure that compares the expected and observed data distributions, where values approaching zero indicate a better fit. F1 Score: A measure of a test's accuracy that balances both precision (the proportion of correct positive predictions) and recall (the proportion of actual positives correctly identified)

[43] applied CV for non-destructive, real-time monitoring of color changes, particularly browning, in organic Gala apples during convective drying with an RMSE below 0.04. The CV-integrated SDS is shown in Fig. 17.

Sampson et al. [90] developed a CV system with two side cameras, high-resolution lenses, and LED illumination to measure color in tray drying. The system used 11 features, correlating with moisture content at an R^2 above 0.9. Udomkun et al. [13] studied online color changes during the hot air drying of papaya using a charge-coupled device (CCD) camera and laser back-scattering, achieving R^2 values over 0.90 for most color attributes.

Shrinkage is another important drying indicator that can affect the food product's quality, appearance, and nutritional value. Excessive shrinkage can lead to a loss of profitability and customer acceptability [91]. Shrinkage occurs during drying when the food contracts into the space cells previously occupied by water. It is determined by changes in the area measured by a caliper or micrometer or by changes in the food microstructure using microscopic techniques. These techniques are tedious and not suitable for online shrinkage measurement. CV technology coupled with ML methods can monitor food shrinkage during drying. Iheonye, Gariepy, and Raghavan [92] used a digital SLR camera to monitor real-time shrinkage of peas during fluidized bed drying, measuring shrinkage based on diameter and area, with excellent performance. Nadian et al. [93] employed CV to estimate shrinkage in kiwifruit during hot air infrared (hybrid) drying, using a genetic algorithm to optimize a fuzzy logic control system, and found that hybrid drying reduced drying load and time. Onwude et al. [94] used a CCD camera with a zoom lens and Image-Pro Insight 9 software in a convective hot-air dryer, applying ANN modeling to accurately predict sweet potato shrinkage and moisture content, achieving correlations above 0.95. CV was also used successfully to determine shrinkage in turmeric slices during solar drying [95] and mushroom air drying [96].

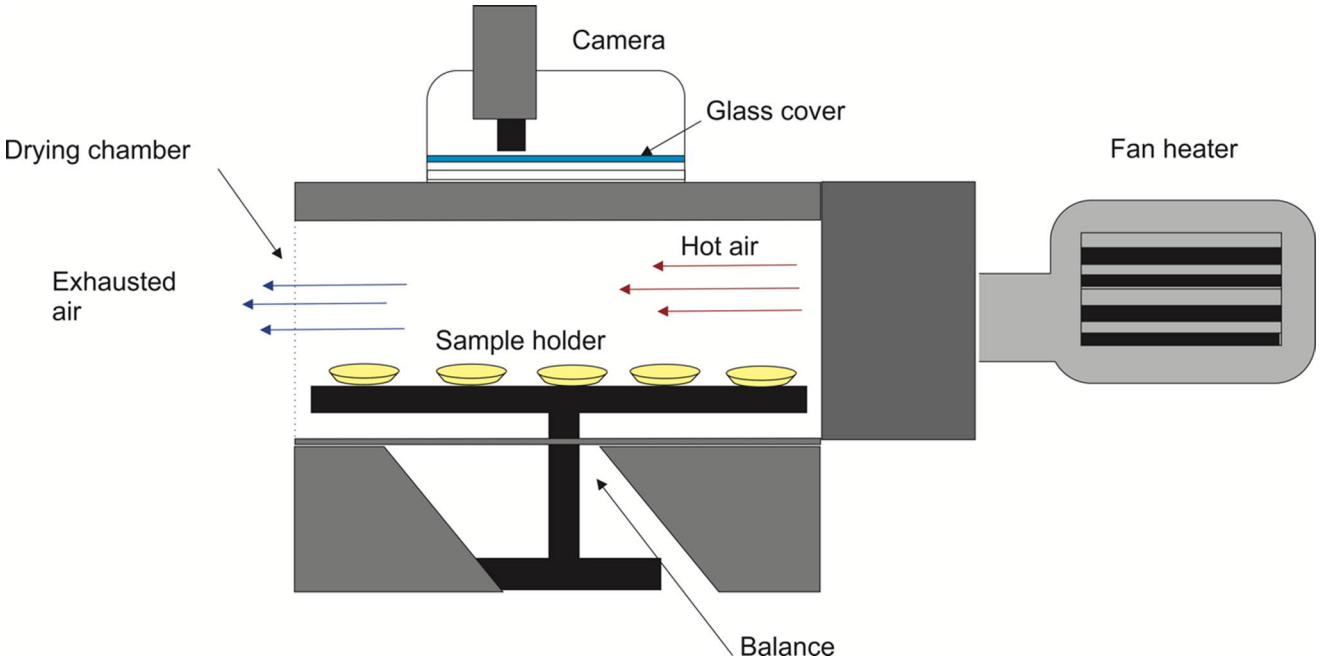


Fig. 17 Schematic of smart dryer convection-based cabinet with CV features

Bulk volume is a key physical property that impacts storage and transportation costs. Rapid and effective measurement of bulk volume during drying is crucial in drying industries. Wang et al. [97] reported an effective CV system for indirect volume measurement during the convective drying of apples. The results showed that the CV system's volume predictions were similar to reference measurements, with only a 6.45% error, indicating that CV is a reliable tool for estimating the bulk volume of food products.

Texture is crucial for consumer acceptability and includes attributes like hardness, chewiness, crispness, creaminess, and smoothness, which are hard to measure during drying using destructive methods. Hosseinpour et al. [98] used a dual-view CV system to measure shrimp texture and employed multilayer perceptron ANN configurations to predict these properties, achieving correlation coefficients above 0.99.

A CV system was tested to differentiate drying techniques for tarhana, a fermented Turkish wheat product. Various ML methods were used to analyze textural and color components from the CV system. The study found that a K-nearest-neighbors classifier using a feature model based on stepwise discriminant analysis achieved the highest accuracy (99.5%) [99]. Determining grain impurities usually involves tedious and inaccurate methods like visual inspection and lab analysis. Recent research showed that a CV system could detect corn contaminants during deep drying [100]. They used the multi-scale-retinex with color restore (MSRCR) technique to reduce image noise, achieving impurity detection rates of 83.05% for broken cobs, 83.87% for broken bracts, and 87.44% for crushed stones. A successful intelligent CV system for the far-infrared drying of ginger slices was reported by Pei et al. [101]. Two fuzzy logic control systems aided by CV efficiently optimized parameters and enhanced product quality by extracting and evaluating image features.

Limitations of Optical Sensing in Smart Drying

Different spectroscopic and imaging approaches, such as NIRS, HSI, and CV, have their limitations when applied in drying systems. RGB imaging captures only three spectral bands (red, green, and blue) in the visible range (400–700 nm), which may not be sufficient for capturing subtle changes in material properties during drying processes. RGB imaging may struggle to differentiate between materials with similar color characteristics, leading to inaccuracy in monitoring drying progress. As RGB imaging is sensitive to lighting conditions, its performance can be affected by variations in lighting conditions that will impact the reliability of real-time monitoring in diverse environments. NIR spectroscopic sensors have limited penetration depth, which may pose challenges in analyzing the internal properties of thick or densely packed materials during drying processes. Although developing NIRS model for the drying process is easy, deploying NIR spectroscopic sensors for real-time monitoring can be costly and requires specialized equipment and expertise. On the other hand, HSI generates large

volumes of data with high spectral resolution, requiring significant computational resources for real-time processing and analysis, which may introduce delays in monitoring drying processes. HSI requires complex calibration procedures to account for variations in illumination, sensor response, and environmental conditions, which can be time-consuming and may introduce errors in the monitoring process. Although HSI provides detailed spectral information, it may have limited spatial resolution compared to other imaging techniques, potentially limiting its ability to capture fine details or small-scale variations during drying processes. Overall, while RGB imaging, NIR spectroscopic sensors, and HSI offer valuable insights into drying processes, their respective limitations must be carefully considered when selecting monitoring techniques for specific applications. Combining multiple techniques or integrating complementary sensors may help overcome some of these limitations and enhance the accuracy and reliability of real-time monitoring in drying applications.

Perspectives of AI-Enabled Smart Drying

For real-time drying process monitoring, the convergence of RGB imaging, NIR spectroscopic sensors, and HSI with AI represents a transformative future. Integrating these technologies overcomes conventional drying process monitoring limitations and propels real-time monitoring capabilities. A complete understanding of drying dynamics can be achieved by synergistically integrating diverse sensor modalities, such as RGB imaging for surface analysis and NIRS for internal property assessment. HSI can be used to understand drying processes completely, encompassing both surface and internal characteristics of materials. Hyperspectral systems will significantly improve spectral resolution, spatial resolution, and data processing capabilities in the coming years. Such integration will enable a smart system to capture detailed information about the composition and properties of drying materials with high accuracy and precision. By leveraging AI, HSI systems can analyze complex datasets in real-time, extract relevant information about moisture content, texture, color, and other key parameters, and provide actionable insights to optimize the drying process. Moreover, miniaturization and cost reduction advancements will make HSI systems more accessible to a wider range of industries and applications. This will facilitate their adoption in various sectors, including food processing, pharmaceuticals, and materials science, where precise control over drying processes is critical for product quality and consistency. Furthermore, developing portable and wearable HSI imaging devices will enable on-site monitoring of drying processes in remote or challenging environments. Such handheld devices will empower industries to implement real-time quality control measures and ensure compliance with regulatory standards, even in the most demanding operating conditions.

AI-driven optical sensing harnesses the power of real-time data mining to translate intricate patterns within vast datasets. Real-time data mining facilitates predictive modeling to anticipate drying outcomes and enables autonomous process control systems to optimize parameters for peak efficiency and quality dynamically. As sensor technologies continue to evolve, with advancements in spectral and spatial resolution, the monitoring will reach new levels of precision. Moreover, the advent of AI-enabled remote monitoring systems promises to extend the reach of monitoring capabilities across different locations, advancing centralized control and coordination for enhanced efficiency and consistency. The ongoing synergy between cutting-edge technologies, industry-university partnerships, and visionary research endeavors is key to modernizing drying operations across many industries.

Conclusions

Smart drying with AI-enabled optical sensing improves product quality, promotes energy efficiency, and enhances sustainability. NIRS, HSI, and CV are promising optical sensing techniques widely applied for real-time monitoring and quality control of dried foods. These techniques allow for non-destructive and rapid measurements, enabling continuous monitoring and timely adjustments to process parameters. NIRS allows real-time, non-destructive moisture content and chemical composition monitoring, enhancing efficiency and product quality. HSI combines the benefits of spectroscopy and imaging, providing detailed information about samples. HSI captures images across narrow, contiguous spectral bands, identifying and quantifying specific chemical components or physical properties, offering comprehensive insights into moisture distribution, structural changes, and chemical composition throughout the drying process. This facilitates the analysis and characterization of product physicochemical and geometrical properties. CV is another smart optical sensing tool for managing drying products, analyzing visual characteristics like color and texture to determine quality and uniformity. Recent studies show that using AI with different optical sensors gives excellent predictive results, leading to smarter, more sustainable, data-driven drying operations. Integrating cutting-edge techniques like NIRS, HSI, and CV with real-time data acquisition and intelligent control algorithms, the food industry can optimize drying processes across

different matrices and systems, enhancing productivity and sustainability.

Acknowledgements This study was financially supported by the Center for Advanced Research in Drying (CARD), a U.S. National Science Foundation Industry University Cooperative Research Center. CARD is located at Worcester Polytechnic Institute and the University of Illinois at Urbana-Champaign (co-site).

Author Contributions M. K. and M.V. d. S. F. conceptualized and designed the manuscript. M.V. d. S. F. wrote the majority of the manuscript text and prepared figures, M.W. A. and M. O. wrote some parts. All authors reviewed, edited and approved the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

References

1. Zielinska M, Markowski M (2012) Color characteristics of carrots: effect of drying and rehydration. Int J Food Prop 15(2):450–466. https://doi.org/10.1080/10942912.2010.489209
2. Calín-Sánchez Á, Lipan L, Cano-Lamadrid M, Kharaghani A, Masztalerz K, Carbonell-Barrachina Á, Figiel A (2020) Comparison of traditional and novel drying techniques and its effect on quality of fruits, vegetables and aromatic herbs. Foods 9(9). https://doi.org/10.3390/foods9091260
3. Vega-Gálvez A, Ah-Hen K, Chacana M, Vergara J, Martínez-Monzó J, García-Segovia P, Lemus-Mondaca R, Di Scala K (2012) Effect of temperature and air velocity on drying kinetics, antioxidant capacity, total phenolic content, colour, texture and microstructure of apple (var. Granny Smith) slices. Food Chem 132(1):51–59. https://doi.org/10.1016/j.foodchem.2011.10.029
4. Ghosh S, Gillis A, Levkov K, Vitkin E, Golberg A (2020) Saving energy on meat air convection drying with pulsed electric field coupled to mechanical press water removal. Innov Food Sci Emerg Technol 66:102509. https://doi.org/10.1016/j.ifset.2020.102509
5. Xing X, Zhang C, Gu J, Zhang Y, Xinrun L, Zhuo Z (2021) Intelligent drying rack system based on internet of things. J Phys: Conf Ser 1887(1). https://doi.org/10.1088/1742-6596/1887/1/012002
6. Menon A, Stojceska V, Tassou SA (2020) A systematic review on the recent advances of the energy efficiency improvements in non-conventional food drying technologies. Trends Food Sci Technol 100:67–76. https://doi.org/10.1016/j.tifs.2020.03.014
7. Moscetti R, Raponi F, Cecchini M, Monarca D, Massantini R (2021) Feasibility of computer vision as process analytical technology tool for the drying of organic apple slices. Acta Horticulturae ISHS 433–438
8. Fracarolli JA, Pavarin FFA, Castro W, Blasco J (2020) Computer vision applied to food and agricultural products. Rev Cienc Agron 51(5):1–20. https://doi.org/10.5935/1806-6690.20200087
9. Lindelauf AA, Saelmans AG, Van Kuijk SMJ, Van der Hulst RWJ, Rutger MS (2022) Near-infrared spectroscopy (NIRS) versus hyperspectral imaging (HSI) to detect flap failure in reconstructive surgery: a systematic review. Life 12(1). https://doi.org/10.3390/life12010065
10. Cho JS, Ji YC, Moon KD (2020) Hyperspectral imaging technology for monitoring of moisture contents of dried persimmons during drying process. Food Sci Biotechnol 29(10):1407–1412. https://doi.org/10.1007/s10068-020-00791-x
11. Netto JMS, Honorato FA, Azoubel PM, Kurozawa LE, Barbin DF (2021) Evaluation of melon drying using hyperspectral imaging technique in the near infrared region. LWT 143:111092. https://doi.org/10.1016/j.lwt.2021.111092
12. Xue H, Xu X, Yang Y, Hu D, Niu G (2024) Rapid and non-destructive prediction of moisture content in maize seeds using hyperspectral imaging. Sensors 24(6):1855. https://doi.org/10.3390/s24061855
13. Udomkun P, Nagle M, Argyropoulos D, Mahayothee B, Müller J (2016) Multi-sensor approach to improve optical monitoring of papaya shrinkage during drying. J Food Eng 189:82–89. https://doi.org/10.1016/j.jfoodeng.2016.05.014
14. Rizwana S, Nesha SF, Kumar K, Pohthmi S, Hazarika MK (2023) Drying kinetics and image-based identification of drying end point during parboiling of Komal Chawal. J Agric Food Res 13:100646. https://doi.org/10.1016/j.jafr.2023.100646
15. Yliniemi L, A Jutila E, P Uronen A (1981) Modelling and control of a pilot plant rotary drier used for drying of industrial ore concentrates. IFAC Proc 14(2):2837–2843. https://doi.org/10.1016/S1474-6670(17)63893-X
16. Sivakumar R, Saravanan R, Perumal AE, Iniyan S (2016) Fluidized bed drying of some agro products– A review. Renew Sustain Energy Rev 61:280–301. https://doi.org/10.1016/j.rser.2016.04.014
17. Oyinloye TM, Yoon WB (2020) Effect of freeze-drying on quality and grinding process of food produce: a review. Processes 8(3):1–23. https://doi.org/10.3390/PR8030354
18. Fernández-Quiroz D, Tohidi MM, Paymard B, Lucero-Acuña A (2023) Immobilization of essential oils in biopolymeric matrices: recent approaches for controlled delivery systems. Stud Nat Prod Chem 78:365–401
19. Ramos FM, Vivaldo SJ, Prata AS (2021) Impact of vacuum spray drying on encapsulation of fish oil: oxidative stability and encapsulation efficiency. Food Res Int 143(March):110283. https://doi.org/10.1016/j.foodres.2021.110283
20. Zielinska M, Michalska A (2016) Microwave-assisted drying of blueberry (*Vaccinium corymbosum* L.) fruits: drying kinetics, polyphenols, anthocyanins, antioxidant capacity, colour and texture. Food Chem 212:671–680. https://doi.org/10.1016/j.foodchem.2016.06.003
21. Khan MKI, Maan AA, Aadil RM, Nazir A, Butt MS, Rashid MI, Afzal MI (2020) Modelling and kinetic study of microwave-assisted drying of ginger and onion with simultaneous extraction of bioactive compounds. Food Sci Biotechnol 29(4):513–519. https://doi.org/10.1007/s10068-019-00695-5
22. Wiktor A, Witrowa-Rajchert D (2020) Drying kinetics and quality of carrots subjected to microwave-assisted drying preceded by combined pulsed electric field and ultrasound treatment. Dry Technol 38(1–2):176–188. https://doi.org/10.1080/07373937.2019.1642347
23. Öztürk F, Gündüz H (2018) The effect of different drying methods on chemical composition, fatty acid, and amino acid profiles of sea cucumber (Holothuria tubulosa Gmelin, 1791). J Food Process Preserv 42(9):1–10. https://doi.org/10.1111/jfpp.13723
24. Vadivambal R, Jayas DS (2011) Applications of thermal imaging in agriculture and food industry—A review. Food Bioproc Technol 4(2):186–199. https://doi.org/10.1007/s11947-010-0333-5
25. Krishnamurthy K, Khurana HK, Soojin J, Irudayaraj J, Demirci A (2008) Infrared heating in food processing: an overview. Compr Rev Food Sci Food Saf 7(1):2–13. https://doi.org/10.1111/j.1541-4337.2007.00024.x

26. Zartha S, Wilder J, Orozco GL, Murillo LMG, Osorio MP, Suarez NS (2021) Infrared drying trends applied to fruit. Front Sustain Food Syst 5:1–14. https://doi.org/10.3389/fsufs.2021.650690
27. Yi Y, Salonitis K, Tsoutsanis P, Litos L, Patsavelas J (2017) Improving the curing cycle time through the numerical modeling of air flow in industrial continuous convection ovens. Procedia CIRP 63:499–504. https://doi.org/10.1016/j.procir.2017.03.167
28. Davidson I (2023) Oven construction: electric ovens. Biscuit Bak Techno 1:201–209. https://doi.org/10.1016/b978-0-323-99923-6.00029-6
29. Prvulovic S, Tolmac D, Lambic M (2007) Convection drying in the food industry. Agric Eng Int: CIGR Ejournal 9:1–13
30. Khatir Z, Paton J, Thompson H, Kapur N, Toropov V, Lawes M, Kirk D (2012) Computational fluid dynamics (CFD) investigation of air flow and temperature distribution in a small-scale bread-baking oven. Appl Energy 89(1):89–96. https://doi.org/10.1016/j.apenergy.2011.02.002
31. Onwude DI, Hashim N, Chen G (2016) Recent advances of novel thermal combined hot air drying of agricultural crops. Trends Food Sci Technol 57:132–145. https://doi.org/10.1016/j.tifs.2016.09.012
32. Mediani A, Hamezah HS, Jam FA, Mahadi NF, Chan SXY, Rohani ER, Che Lah NH, Azlan UK, Khairul Annuar NA, Azman NAF, Bunawan H, Sarian MN, Kamal N, Abas F (2022) A comprehensive review of drying meat products and the associated effects and changes. Front Nutr 9:1–24. https://doi.org/10.3389/fnut.2022.1057366
33. Su Y, Zhang M, Mujumdar AS (2015) Recent developments in smart drying technology. Dry Technol 33(3):260–276. https://doi.org/10.1080/07373937.2014.985382
34. Sun Q, Zhang M, Mujumdar AS (2019) Recent developments of artificial intelligence in drying of fresh food: a review. Crit Rev Food Sci Nutr 59(14):2258–2275. https://doi.org/10.1080/10408398.2018.1446900
35. Fan S, Zhang B, Li J, Huang W, Wang C (2016) Effect of spectrum measurement position variation on the robustness of NIR spectroscopy models for soluble solids content of apple. Biosyst Eng 143:9–19. https://doi.org/10.1016/j.biosystemseng.2015.12.012
36. Tsenkova R, Muncan J, Pollner B, Kovacs Z (2018) Essentials of aquaphotomics and its chemometrics approaches. Front Chem 6:1–25. https://doi.org/10.3389/fchem.2018.00363
37. Maduro Dias CSA, Nunes HP, Melo TMMV, Rosa HJD, Silva CCG, Borba AES (2021) Application of near infrared reflectance (NIR) spectroscopy to predict the moisture, protein, and fat content of beef for gourmet hamburger preparation. Livest Sci 254:104772. https://doi.org/10.1016/j.livsci.2021.104772
38. Pasquini C (2018) Near infrared spectroscopy: a mature analytical technique with new perspectives– A review. Anal Chim Acta 1026:8–36. https://doi.org/10.1016/j.aca.2018.04.004
39. Huang L, Zhao J, Chen Q, Zhang Y (2014) Nondestructive measurement of total volatile basic nitrogen (TVB-N) in pork meat by integrating near infrared spectroscopy, computer vision and electronic nose techniques. Food Chem 145:228–236. https://doi.org/10.1016/j.foodchem.2013.06.073
40. Kamruzzaman M, Sun D-W (2016) Introduction to hyperspectral imaging technology. In: Sun D-W (ed) Computer vision technology for food quality evaluation, 2nd edn. Academic, pp 111–139. https://doi.org/10.1016/B978-0-12-802232-0.00005-0
41. Ahmed MW, Hossainy SJ, Khaliduzzaman A, Emmert JL, Kamruzzaman M (2023) Non-destructive optical sensing technologies for advancing the egg industry toward industry 4.0: a review. Compr Rev Food Sci Food Saf 22:4378–4403. https://doi.org/10.1111/1541-4337.13227
42. Chakravartula SSN, Bandiera A, Nardella M, Bedini G, Ibba P, Massantini R, Moscetti R (2023) Computer vision-based smart monitoring and control system for food drying: a study on carrot slices. Comput Electron Agric 206:107654. https://doi.org/10.1016/j.compag.2023.107654
43. Raponi F, Moscetti R, Chakravartula SSN, Fidaleo M, Massantini R (2022) Monitoring the hot-air drying process of organically grown apples (cv. Gala) using computer vision. Biosyst Eng 223:1–13. https://doi.org/10.1016/j.biosystemseng.2021.07.005
44. Xiao Z, Wang J, Han L, Guo S, Cui Q (2022) Application of machine vision system in food detection. Front Nutr 9:1–7. https://doi.org/10.3389/fnut.2022.888245
45. Zhang L, Zhang M, Mujumdar AS (2023) Terahertz spectroscopy: a powerful technique for food drying research. Food Rev Int 39(3):1733–1750. https://doi.org/10.1080/87559129.2021.1936004
46. Khan PW, Yung-Cheol B, Park N (2020) IoT-blockchain enabled optimized provenance system for food industry 4.0 using advanced deep learning. Sensors 20(10). https://doi.org/10.3390/s20102990
47. Kushwah A, Gaur Mk, Kumar A, Singh P (2022) Application of ANN and prediction of drying behavior of mushroom drying in side hybrid greenhouse solar dryer: an experimental validation. J Therm Eng 8(2):221–234. https://doi.org/10.18186/thermal.1086189
48. Zhu L, Spachos P, Pensini E, Plataniotis KN (2021) Deep learning and machine vision for food processing: a survey. Curr Res Food Sci 4:233–249. https://doi.org/10.1016/j.crfs.2021.03.009
49. Santhi R, Abirami, Muthuswamy P (2023) Industry 5.0 or industry 4.0S? Introduction to industry 4.0 and a peek into the prospective industry 5.0 technologies. Int J Interact Des Manuf 17(2):947–979. https://doi.org/10.1007/s12008-023-01217-8
50. Gowen AA, O'Donnell CP, Cullen PJ, Downey G, Frias JM (2007) Hyperspectral imaging - an emerging process analytical tool for food quality and safety control. Trends Food Sci Technol 18(12):590–598. https://doi.org/10.1016/j.tifs.2007.06.001
51. Khan MIH, Joardder MUH, Karim MA (2022) Application of machine learning-based approach in food drying: opportunities and challenges. Dry Technol 40(6):1051–1067. https://doi.org/10.1080/07373937.2020.1853152
52. Merino A, Alves R, Acebes LF, Prada C (2017) Modeling and simulation of a beet pulp dryer for a training simulator. Dry Technol 35(14):1765–1780. https://doi.org/10.1080/07373937.2016.1275674
53. Çakmak G, Yildiz C (2011) The prediction of seedy grape drying rate using a neural network method. Comput Electron Agric 75(1):132–138. https://doi.org/10.1016/j.compag.2010.10.008
54. Leys L, Nuytten G, Lammens J, Pieter-Jan VB, Corver J, Vervaet C, De Beer T (2021) A NIR-based study of desorption kinetics during continuous spin freeze-drying. Pharmaceutics 13(12):2168. https://doi.org/10.3390/pharmaceutics13122168
55. Parikh D (2014) Solids drying: basics and applications. Chem Eng 1:42–45
56. Ishikawa D, Ueno G, Fujii T (2017) Estimation method of moisture content at the meat surface during drying process by NIR spectroscopy and its application for monitoring of water activity. Jpn J Food Eng 18(3):135–143. https://doi.org/10.11301/jsfe.17493
57. Achata EM, Esquerre C, Shikha KO, Tiwari BK, O'Donnell CP (2021) Development of NIR-HSI and chemometrics process analytical technology for drying of beef jerky. Innov Food Sci Emerg Technol 69:102611. https://doi.org/10.1016/j.ifset.2021.102611
58. Amjad W, Munir A, Sturm B (2020) Development of an imaging system for spatially real-time measurement of drying parameters in industrial drying units. Agric Eng Int: CIGR J 22(4):238–249
59. Collell C, Gou P, Arnau J, Comaposada J (2011) Non-destructive estimation of moisture, water activity and NaCl at ham surface during resting and drying using NIR spectroscopy. Food Chem 129(2):601–607. https://doi.org/10.1016/j.foodchem.2011.04.073

60. Sinelli N, Casiraghi E, Barzaghi S, Brambilla A, Giovanelli G (2011) Near infrared (NIR) spectroscopy as a tool for monitoring blueberry osmo–air dehydration process. Food Res Int 44(5):1427–1433. https://doi.org/10.1016/j.foodres.2011.02.046
61. Mensink MA, Bockstal PJV, Pieters S, Laurens DM, Frijlink HW, Maarschalk KV, Voort HJL, De Beer T (2015) In-line near infrared spectroscopy during freeze-drying as a tool to measure efficiency of hydrogen bond formation between protein and sugar, predictive of protein storage stability. Int J Pharm 496(2):792–800. https://doi.org/10.1016/j.ijpharm.2015.11.030
62. Campos MI, Mussons ML, Antolin G, Debán L, Pardo R (2017) On-line prediction of sodium content in vacuum packed dry-cured ham slices by non-invasive near infrared spectroscopy. Meat Sci 126:29–35. https://doi.org/10.1016/j.meatsci.2016.12.005
63. Moscetti R, Flavio R, Ferri S, Colantoni A, Monarca D, Massantini R (2018) Real-time monitoring of organic apple (var. Gala) during hot-air drying using near-infrared spectroscopy. J Food Eng 222:139–150. https://doi.org/10.1016/j.jfoodeng.2017.11.023
64. Phetpan K, Udompetaikul V, Sirisomboon P (2019) In-line near infrared spectroscopy for the prediction of moisture content in the tapioca starch drying process. Powder Technol 345:608–615. https://doi.org/10.1016/j.powtec.2019.01.050
65. Liu W, Zhang M, Bhandari B, Yu D (2021) A novel combination of LF-NMR and NIR to intelligent control in pulse-spouted microwave freeze drying of blueberry. LWT 137:110455. https://doi.org/10.1016/j.lwt.2020.110455
66. Sturm B, Moscetti R, Crichton SOJ, Raut S, Bantle M, Massantini R (2019) Feasibility of Vis/NIR spectroscopy and image analysis as basis of the development of smart-drying technologies. In IDS 2018–21st International Drying Symposium, València, Spain, 9:11–14. https://doi.org/10.4995/ids2018.2018.7616
67. Cozzolino D, Phan ADT, Netzel M, Smyth H, Sultanbawa Y (2021) Assessing the interaction between drying and addition of maltodextrin to Kakadu plum powder samples by two dimensional and near infrared spectroscopy. Spectrochim Acta A 247:119121. https://doi.org/10.1016/j.saa.2020.119121
68. Malvandi A, Feng H, Kamruzzaman M (2022) Application of NIR spectroscopy and multivariate analysis for non-destructive evaluation of apple moisture content during ultrasonic drying. Spectrochim Acta A 269:120733. https://doi.org/10.1016/j.saa.2021.120733
69. Kapoor R, Malvandi A, Feng H, Kamruzzaman M (2022) Real-time moisture monitoring of edible coated apple chips during hot air drying using miniature NIR spectroscopy and chemometrics. LWT 154:112602. https://doi.org/10.1016/j.lwt.2021.112602
70. Ferreira MVS, Malvandi A, Lee Y, Kamruzzaman M (2023) Advanced AI-powered drying system with near-infrared assistance for precise endpoint prediction in apple slice drying (Report Q4). Center for Advanced Research in Drying (CARD), Urbana-Champaign/Washington, US
71. Jin W, Zhang M, Mujumdar AS et al (2024) Application of portable NIR spectroscopy for instant prediction of the product quality of apple slices during hot air-assisted radio frequency drying. Food Bioprocess Technol. https://doi-org.ez30.periodicos.capes.gov.br/https://doi.org/10.1007/s11947-024-03343-x
72. Yu P, Huang M, Zhang M, Yang B (2019) Optimal wavelength selection for hyperspectral imaging evaluation on vegetable soybean moisture content during drying. Appl Sci 9(2):331. https://doi.org/10.3390/app9020331
73. Pu Y-Y, Sun D-W (2016) Prediction of moisture content uniformity of microwave-vacuum dried mangoes as affected by different shapes using NIR hyperspectral imaging. Innov Food Sci Emerg Technol 33:348–356. https://doi.org/10.1016/j.ifset.2015.11.003
74. Amjad W, Crichton SOJ, Munir A, Hensel O, Sturm B (2018) Hyperspectral imaging for the determination of potato slice moisture content and chromaticity during the convective hot air drying process. Biosyst Eng 166:170–183. https://doi.org/10.1016/j.biosystemseng.2017.12.001
75. Ren Y, Sun D-W (2022) Monitoring of moisture contents and rehydration rates of microwave vacuum and hot air dehydrated beef slices and splits using hyperspectral imaging. Food Chem 382:132346. https://doi.org/10.1016/j.foodchem.2022.132346
76. Saleh RM, Kulig B, Arefi A, Hensel O, Sturm B (2022) Prediction of total carotenoids, color, and moisture content of carrot slices during hot air drying using non-invasive hyperspectral imaging technique. J Food Process Preserv 46(9):1–20. https://doi.org/10.1111/jfpp.16460
77. Ndisya J, Gitau A, Mbuge D, Arefi A, Bădulescu L, Pawelzik E, Hensel O, Sturm B (2021) Vis-NIR hyperspectral imaging for online quality evaluation during food processing: a case study of hot air drying of purple-speckled cocoyam (Colocasia esculenta (L.) Schott). Processes 9(10). https://doi.org/10.3390/pr9101804
78. Crichton S, Shrestha L, Hurlbert A, Sturm B (2017) Use of hyperspectral imaging for the prediction of moisture content and chromaticity of raw and pretreated apple slices during convection drying. Dry Technol 36(7):804–816. https://doi.org/10.1080/07373937.2017.1356847
79. Liu Y, Sun Y, Xie A, Yu H, Yin Y, Li X, Duan X (2017) Potential of hyperspectral imaging for rapid prediction of anthocyanin content of purple-fleshed sweet potato slices during drying process. Food Anal Methods 10(12):3836–3846. https://doi.org/10.1007/s12161-017-0950-y
80. Lin LX, Xu J-L, Sun D-W (2020) Evaluating drying feature differences between ginger slices and splits during microwave-vacuum drying by hyperspectral imaging technique. Food Chem 332:127407. https://doi.org/10.1016/j.foodchem.2020.127407
81. Chen X, Jiao Y, Liu B, Chao W, Duan X, Yue T (2022) Using hyperspectral imaging technology for assessing internal quality parameters of persimmon fruits during the drying process. Food Chem 386:132774. https://doi.org/10.1016/j.foodchem.2022.132774
82. Yang Q, Sun D-W, Cheng W (2017) Development of simplified models for nondestructive hyperspectral imaging monitoring of TVB-N contents in cured meat during drying process. J Food Eng 192:53–60. https://doi.org/10.1016/j.jfoodeng.2016.07.015
83. Arefi A, Sturm B, Hensel O, Raut S (2023) NIR monochrome imaging for monitoring of apple drying process: light-emitting diode and band-pass filter imaging techniques. Food Biosci 54. https://doi.org/10.1016/j.fbio.2023.102898. 102 Phetpan 8
84. Zubkova K, Sherstjuk V (2023) Neuro-fuzzy control of spray drying food machine. CEUR Workshop Proc 3373:129–145
85. Zareiforoush H, Minaei S, Alizadeh MR, Banakar A (2015) Potential applications of computer vision in quality inspection of rice: a review. Food Eng Rev 7(3):321–345. https://doi.org/10.1007/s12393-014-9101-z
86. Bantle M, Christian AK, Claussen IC, Tolstorebrov I (2019) Influence of the low temperature drying process on optical alternations of organic apple slices. Proceedings of the 21st International Drying Symposium 11–14. https://doi.org/10.4995/ids2018.2018.7551
87. Moscetti R, Chakravartula SSN, Bandiera A, Bedini G, Massantini R (2020) Computer vision technology for quality monitoring in smart drying system. IEEE Int Workshop Metrol Agric for. 134–138
88. Hosseinpour S, Rafiee S, Mohtasebi SS, Aghbashlo M (2013) Application of computer vision technique for on-line monitoring of shrimp color changes during drying. J Food Eng 115(1):99–114. https://doi.org/10.1016/j.jfoodeng.2012.10.003
89. Onwude DI, Hashim N, Abdan K, Janius R, Chen G (2018a) Combination of computer vision and backscattering imaging for predicting the moisture content and colour changes of sweet potato (Ipomoea batatas L.) during drying. Comput Electron Agric 150:178–187. https://doi.org/10.1016/j.compag.2018.04.015

90. Sampson DJ, Chang YK, Rupasinghe HPV, Zaman QUZ (2014) A dual-view computer-vision system for volume and image texture analysis in multiple apple slices drying. J Food Eng 127:49–57. https://doi.org/10.1016/j.jfoodeng.2013.11.016
91. Zhang W, Liu T, Ueland M, Forbes SL, Wang RX, Su SW (2020) Design of an efficient electronic nose system for odour analysis and assessment. Meas: J Int Meas Confed 165:108089. https://doi.org/10.1016/j.measurement.2020.108089
92. Iheonye A, Gariepy Y, Raghavan V (2020) Computer vision for real-time monitoring of shrinkage for peas dried in a fluidized bed dryer. Dry Technol 38(1–2):130–146. https://doi.org/10.1080/07373937.2019.1649277
93. Nadian MH, Abbaspour-Fard MH, Martynenko A, Golzarian MR (2017) An intelligent integrated control of hybrid hot air-infrared dryer based on fuzzy logic and computer vision system. Comput Electron Agric 137:138–149. https://doi.org/10.1016/j.compag.2017.04.001
94. Onwude DI, Hashim N, Abdan K, Janius R, Chen G (2018b) The potential of computer vision, optical backscattering parameters and artificial neural network modelling in monitoring the shrinkage of sweet potato (Ipomoea batatas L.) during drying. J Sci Food Agric 98(4):1310–1324. https://doi.org/10.1002/jsfa.8595
95. Sharma S, Dhalsamant K, Tripathy PP (2019) Application of computer vision technique for physical quality monitoring of turmeric slices during direct solar drying. J Food Meas Charact 13(1):545–558. https://doi.org/10.1007/s11694-018-9968-0
96. Li X, Liu Y, Gao Z, Xie Y, Wang H (2021) Computer vision online measurement of shiitake mushroom (Lentinus edodes) surface wrinkling and shrinkage during hot air drying with humidity control. J Food Eng 292:110253. https://doi.org/10.1016/j.jfoodeng.2020.110253
97. Wang D, Martynenko A, Corscadden K, He Q (2017) Computer vision for bulk volume estimation of apple slices during drying. Dry Technol 35(5):616–624. https://doi.org/10.1080/07373937.2016.1196700
98. Hosseinpour S, Rafiee S, Aghbashlo M, Mohtasebi SS (2015) Computer vision system (CVS) for in-line monitoring of visual texture kinetics during shrimp (Penaeus spp.) drying. Drying Technol 33(2):238–254. https://doi.org/10.1080/07373937.2014.947513
99. Kurtulm S, Rafiee S, Mohtasebi SS, Aghbashlo M (2013) Application of computer vision technique for on-line monitoring of shrimp color changes during drying. J Food Eng 115(1):99–114. https://doi.org/10.1016/j.jfoodeng.2012.10.003
100. Li T, Tong J, Liu M, Yao M, Xiao Z, Li C (2022) Online detection of impurities in corn deep-bed drying process utilizing machine vision. Foods 11(24). https://doi.org/10.3390/foods11244009
101. Pei Y, Li Z, Ling C, Jiang L, Wu X, Song C, Li J, Song F, Xu W (2022) An improvement of far-infrared drying for ginger slices with computer vision and fuzzy logic control. J Food Meas Charact 16(5):3815–3831. https://doi.org/10.1007/s11694-022-01453-8
102. Muruganantham P, Samrat NH, Islam N, Johnson J, Wibowo S, Grandhi S (2023) Rapid estimation of moisture content in unpeeled potato tubers using hyperspectral imaging. Appl Sci 13(1). https://doi.org/10.3390/app13010053
103. Pu Y-Y, Sun D-W (2015) Vis–NIR hyperspectral imaging in visualizing moisture distribution of mango slices during microwave-vacuum drying. Food Chem 188:271–278. https://doi.org/10.1016/j.foodchem.2015.04.120
104. Xu W, Zhang F, Wang J, Ma Q, Sun J, Tang Y, Wang J, Wang W (2022) Real-time monitoring of the quality changes in shrimp (*Penaeus vannamei*) with hyperspectral imaging technology during hot air drying. Foods 11(20). https://doi.org/10.3390/foods11203179
105. Przybył K, Gawałek J, Koszela K (2023) Application of artificial neural network for the quality-based classification of spray-dried rhubarb juice powders. J Food Sci Technol 60(3):809–819. https://doi.org/10.1007/s13197-020-04537-9
106. Kurtulmuş F, Gürbüz O, Değirmencioğlu N (2014) Discriminating drying method of tarhana using computer vision. J Food Process Eng 37(4):362–374. https://doi.org/10.1111/jfpe.12092

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.